

Information Access Speed and Organizational Performance*

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Abstract

Recent technological advancements have enabled more data-driven decision-making across firms. While prior literature highlights the value of using more data, there is less insight on the impact of information *speed*, particularly when decision-makers access it. We study how faster access influences organizational decisions, using data from healthcare. We examine a technology that displayed test results in real time, accelerating information access for 64,152 decisions by 387 physicians. Faster access not only expedited decisions but also allowed decision-makers to gather subsequent information more efficiently and improve outcomes, boosting organizational performance. Our findings show that investing in data velocity can yield significant benefits through both faster and better decisions.

Keywords: Data-driven decision-making, experimentation, technology, speed, information, healthcare

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1 Introduction

Recent technological advancements have enabled new opportunities to collect and leverage data across the organization, providing firms with more data to inform their decisions than ever before. This has led to changes in how managers and firms make decisions, leading them to rely more on data rather than intuition (Brynjolfsson and McElheran, 2016, 2019; Adner et al., 2019). Much recent work has highlighted the value of information for decision-making, showing that the availability and use of data improves firm decisions (Agrawal et al., 2019; Camuffo et al., 2020; Koning et al., 2022), and plays a role in driving performance differences across small and large firms alike (Nagaraj, 2022; Kim, 2024; Agarwal et al., 2023; Galdon-Sanchez et al., 2024).

While much focus has been placed on the *amount* of information and its value, another key implication of recent technological changes is the change in the *speed* of information: firms increasingly have access to real-time data to inform their decisions, and this increased speed may change how they make decisions. Firms invest in this data velocity by increasing the speed of data acquisition, transmission, or processing – including the speed at which relevant insights are accessed and attended to by decision-makers, which ultimately determines whether any available data translates into strategic advantage.

However, there has been less insight into how data velocity affects decisions in organizations. While we might expect that it should lead to *faster* decisions, how it affects the efficiency of subsequent information acquisition and *quality* of decision-making is far more ambiguous. On the one hand, it may have little effect, leading to the same decisions simply made faster. On the other hand, it might change the decisions themselves and how they are made by altering what information decision-makers decide to acquire in that process.

In this paper, we propose that increasing the speed of information access results in faster, more efficient, and higher-quality decisions, and provide empirical evidence from a healthcare context. We find that the speed of information access not only increases the speed of decisions but also enables a more efficient decision-making process that leads to the acquisition of less but more targeted

information. Furthermore, this increase in information access speed enhances decision quality, leading to improved indicators of organizational performance. Our findings thus suggest that the speed of data access may generate advantages from faster, efficient, and better firm decisions. This has implications for research on experimentation in decision-making, suggesting that investing in information access speed may help decision-makers improve their theories and develop better experiments (Camuffo et al., 2023; Agrawal et al., 2024; Camuffo et al., 2020). It also provides insights on the broader role of information access speed in building competitive advantages based on speed (Eisenhardt, 1989; Baum and Wally, 2003; Hawk et al., 2013). More broadly, these findings highlight the importance of understanding not only what decisions are made but *how* they are made within organizations, especially in the age of data.

We examine the impact of a technology intervention that increased the speed of information access across 64,152 decisions made by 387 physicians serving 43,607 unique patients on *non-critical* cases in an Emergency Department (ED) of a major hospital.¹ This provided us with uniquely rich and detailed administrative data on a large number of decisions across a large number of decision-makers, and allowed us to observe the entire process of information acquisition (e.g., the tests they order), how quickly they make their decisions based on this information (e.g., the length of time taken), and the performance outcomes of this decision for the organization (e.g., the charges billed to the insurer, the quality of care, and patient satisfaction). The technology was installed in only one of the two adult wards in the hospital at a moment in time when no other changes were introduced in the organization, permitting the use of a difference-in-differences strategy (Chan, 2016).

We leverage a triple-differences strategy that additionally controls for potential seasonal effects to estimate the effect of this technology, which increased the speed of information provided to physicians. Before its implementation, a physician in our sample ordering a laboratory test for a

¹Importantly, all critical cases arriving by ambulance requiring immediate physician attention (i.e., those with high mortality risk) were dispatched to a separate ward in this hospital (resuscitation room). In such cases, the relationship between speed and organizational performance may be more directly linked and in part mechanical. We do not study this ward in this paper.

patient was not automatically notified of test results when they became available.² Instead, the physician had to navigate several screens in the internal software system to check manually whether the result was ready. This process required entering the physician’s password and could not be delegated to a nurse. This friction meant that physicians did not check frequently to save time, a uniquely scarce resource in this setting, and thus experienced long lags in receiving information on the tests they ordered. In June 2022, a dashboard was installed in one ward, which visually notified physicians when test results were ready and whether critical, thereby increasing the speed of information access provided for the decisions they made.³

We find that the intervention increased the speed of decisions. Its introduction was associated with a significant decrease (13% in our most conservative estimate; approximately 103 minutes) in the length of stay of the average patient, which we interpret as indicating an increase in the speed of decisions. A leads and lags exercise indicates that the average length of stay evolved broadly similarly across the two wards in the months before June 2022, and then discontinuously decreased in the treated ward coinciding with the introduction of the technology.

We also find that this had broader effects on organizational outcomes, suggesting that decisions became more efficient and higher-quality. We observe that this improved the efficiency of decision-making, leading to fewer, more targeted tests being ordered and resulting in a 25% decrease in the total charges billed (estimated at around 365 US dollars, almost 1.5x the local monthly minimum wage in 2022). Despite faster and more efficient decisions, we find that the introduction of the technology also reduced patient hospitalization, without increasing the likelihood that the patient returned to the ED within 30 days. To the extent that these variables proxy for the health outcome

²During the entire sample period both pre- and post-intervention, the laboratory called the physicians in both wards only in the rare cases when a lab test result fell outside the normal range, and in those cases, they also communicated the exact value of the result.

³For each laboratory test ordered, the system displayed its processing status, notifying physicians whether the result was ready and whether it fell within pre-established reference ranges. These notifications were visible to all physicians and nurses in the ward but did not include specific test results, physician names, or patient names—only the patient’s national ID number. Importantly, no active alerts or push notifications were sent to any physician devices; instead, physicians had to log into the internal system to manually view the test results. In addition to signaling the availability of results, the technology also indicated the severity of those results. While this feature could theoretically influence physicians’ strategic decision-making—such as prioritizing which patient to see first—we believe this effect is limited. Physicians were already informed, both before and after the introduction of the technology, when a test result fell within a critical range requiring urgent attention.

of the patient, we conclude that the technology pareto-improved the quality of care for patients – suggesting that physicians made better decisions. Using responses from a random sample of patients to a survey conducted by the hospital, we also find an improvement in patient satisfaction.

Contribution to the literature We contribute to two strands of literature. First is the growing body of research on data-driven decision-making, which has provided insight on the value of information for decisions (Brynjolfsson and McElheran, 2019; Camuffo et al., 2020; Koning et al., 2022; Nagaraj, 2022; Kim, 2024; Kim et al., 2024). Our findings suggest that in addition to the amount of data, data velocity – especially in terms of the speed of information access – is an important determinant of how decision-makers learn to improve strategic decisions. Our study also highlights a key mechanism that determines how decision-makers learn to improve strategic decisions: their choice of what information to acquire and how.

Second, our paper contributes to research on speed or time as a source of competitive advantage. Much work has emphasized the importance of speed for organizational performance and competitive advantage (Stalk, 1988; Eisenhardt, 1989; Stalk Jr and Hout, 1990; Forbes, 2005; Hawk et al., 2013; Teece et al., 1997; Helfat et al., 2009), through a first-mover advantage (Makadok, 1998), a series of temporary advantages (Garud et al., 1998), or improved learning via more interactions with the environment (Eisenhardt, 1989; Mosakowski, 1997). This literature has generally conceptualized speed as an advantage in competition, for example by enabling firms to more rapidly respond to customer demand and realize revenues from investments relative to competitors (Stalk, 1988; Hawk et al., 2013; Pacheco-de Almeida et al., 2015) – a key exception being Eisenhardt (1989), which focuses on how speed affects decisions. Much of this literature has also been theoretical, qualitative, or correlational – providing valuable insights but making it difficult to disentangle the impact of speed from other related changes such as flexibility (e.g., just-in-time production) or better management practices. In this paper, we study this phenomenon deeply across thousands of decisions within an organization and highlight a novel lever through which organizations can increase decision speed – improving the speed of information access. We also provide evidence that

suggests that the speed of information access *causally* enables better and more efficient decisions, leading to improved organizational performance.

2 Theoretical framework

In this section, we examine how information shapes firm decisions and performance, with particular emphasis on the speed at which decision-makers access relevant information. We contend that in addition to the quantity of information, the speed of information plays a key role in firm decisions. We then separate the components of information speed, drawing attention to the speed of information access as a pivotal yet underexplored driver of firm decisions and outcomes. Finally, we present our hypotheses on how the speed of information access affects the speed and quality of decisions in organizations.

2.1 The role of information in firm decisions and performance

Information – on competitors, customers, suppliers, and internal operations – is a crucial input to firm decisions. From the literature on evidence-based management (Pfeffer and Sutton, 2006; Barends and Rousseau, 2018) to more recent work on data-driven decision-making (Brynjolfsson and McElheran, 2019), growing research has shown that using information and data-driven processes is a key driver of firm decisions and performance differences across firms in entrepreneurial (Camuffo et al., 2020; Koning et al., 2022), innovation (Nagaraj, 2022), and competitive strategy contexts (Kim, 2024). Complex decisions in organizations require decision-making under uncertainty, which data can help inform (Agrawal et al., 2019; Kim et al., 2024).

Moreover, research on experimentation has highlighted that taking actions to acquire information is a core driver of how organizations learn to make better decisions (Ries, 2011; Eisenmann et al., 2012; Blank, 2013; Kerr et al., 2014; Gans et al., 2019; Camuffo et al., 2020; Leatherbee and Katila, 2020; Koning et al., 2022). Although much of this work has focused on the entrepreneurial context, recent approaches in strategic decision-making have highlighted parallels outside of en-

trepreneurship. Applying a real options approach to strategy emphasizes how initial investments allow firms to collect signals about possible options (Adner and Levinthal, 2004). A Mendelian view of how executives operate highlights the role of intentionality in generating options, testing them through “experiments”, and making a choice (Levinthal, 2017). Across this literature, including in entrepreneurial strategy, experiments generally refer not to formal experimentation, but more broadly to actions designed to yield the gathering of information to inform a choice (Kim, 2025). For example, this may entail interviewing customers or releasing a prototype, as well as launching an A/B test.

However, much of this research has focused on the value of acquiring *more* information for decisions (Koning et al., 2022; Nagaraj, 2022; Kim, 2024), rather than on *how* decision-makers choose to acquire information. Research on evidence-based and data-driven decision-making has primarily focused on the availability of data and its use (e.g., Brynjolfsson and McElheran (2016)), largely providing evidence that more information can be valuable. Recent work has provided additional nuance, highlighting how early information can cast a “streetlight effect”, leading to suboptimal decisions in innovation (Hoelzemann et al., 2024), at times by increasing cognitive overload (Deck and Jahedi, 2015).

It has yielded less insight on key decisions that organizations make on *how* to acquire information, such as the timing of information access or the validity of the method chosen. Similarly, research on experimentation has generally focused on whether information is acquired as a binary action and emphasized the importance of complements such as specifying theories (Camuffo et al., 2020; Agarwal et al., 2023), leaving the design of the actual experimentation process of information acquisition as a black box (Kim, 2025).⁴

⁴Recent work by Gans (2023) pioneers exploration into this question by highlighting different types of information that firms can gather, focusing on whether they set a high- or low-bar on how likely it is to lead to a positive signal regarding a technology’s potential.

2.2 Speed of information access as a key organizational lever

In particular, a key dimension that has been overlooked is the *speed* or *velocity* of information. This is somewhat surprising, given that the excitement around “big data” in management highlights three defining attributes: the “volume” of available data, the “velocity” of data creation, and the “variety” of data types that organizations can use (McAfee et al., 2012). However, while research has extensively examined how volume and variety of data impact firm decisions, the strategic implications of information velocity remain underexplored.

Although these attributes can often come as a bundle, data velocity is distinct from data availability or data acquisition costs, which have been extensively studied both theoretically and empirically (see Bloom et al. (2014) and Goldfarb and Tucker (2019) for reviews). Although data velocity is frequently discussed, it is often in broad terms, rather than precisely defined or conceptualized. We distinguish three constituent components of data velocity: the speed of acquiring, transmitting, and processing data (Chen et al., 2012; Bharadwaj et al., 2013; Davenport and Patil, 2012). Organizations can invest in any of these components to increase data velocity. The speed of data acquisition can be increased by generating or capturing data more quickly, such as deploying sensors in retail environments to capture customer behavior in real time (McAfee et al., 2012). The speed of data transmission refers to the technical efficiency of moving data through networks, and can be enhanced through investments in infrastructure that reduce latency. For instance, financial firms invest in high-speed connections to reduce market data latency (Brogaard et al., 2014). Finally, the speed of data processing can be improved in two ways: (1) by increasing the speed of transforming raw data into insights, and (2) by increasing the speed at which relevant insights are accessed and attended to by decision-makers. For example, firms can implement edge computing to analyze streaming data in near real-time (Shi et al., 2016), or adopt dashboards, sensors, and alert systems that ensure faster delivery of insights to decision-makers.

Naturally, investing in any of these components of data velocity has been seen to enable agility in decisions relative to competitors, providing insights on the market before others make a move

(McAfee et al., 2012). For example, hedge funds and investment banks use satellite data to count cars in retailers’ car parks ahead of corporate earnings releases, detect oil storage tanks across the world to gauge movements in oil prices before they happen, or identify events like hurricanes or wildfires before reports (Hawser, 2022). Similarly, many businesses highlight how processing customer data in real time enable them to respond more quickly to customers and identify changes in demand (Garduno, 2022).

However, the speed of information access—how quickly decision-makers can retrieve and leverage relevant information for their decisions—ultimately determines whether data translates into strategic advantage. Organizations may excel at collecting and transmitting information, yet fail strategically if decision-makers cannot access it when needed (Kim, 2024). Real-time satellite data, for instance, offers little strategic value if executives cannot incorporate it into their decision-making with comparable speed, whether manually or via algorithms. This critical bottleneck—the time between information availability and accessibility to decision-makers—defines whether organizations capitalize on their information capabilities. In fact, speed of information access can lead to temporal advantages that persist even when all competitors eventually collect and process the same information.

2.3 How the speed of information access impacts decisions

Nuances in how the amount of information affects decisions raise the question of whether and how faster access might impact decision speed. Prior work has shown that more information can lead to disregard of additional data (Allen and McDonald, 2025), cognitive overload (Deck and Jahedi, 2015), and suboptimal updating (Hoelzemann et al., 2024). This raises the possibility that faster information access might not result in faster decisions. Nevertheless, holding the amount of information constant, the primary reason proposed for why organizations invest in information speed is that this can lead to speed-based advantages. This insight is highlighted in the literature on speed and competitive advantage, which suggests that executives who can access information quickly can make faster decisions (Eisenhardt, 1989). This research suggests that reaching the

same decision simply faster may enable firms to exploit opportunities and capture customers before their competitors (Hawk et al., 2013; Pacheco-de Almeida et al., 2015; Stalk, 1988), strengthen commitment from their stakeholders (Langley, 1995; Pfeffer and Sutton, 2000), and learn more quickly by making more decisions (Eisenhardt, 1989). While much of this literature has been theoretical, correlational, or qualitative in nature, with some cases yielding contradictory results (e.g., Perlow et al. (2002)), it suggests that accessing information more quickly may lead to decision speed.

This leads us to our first hypothesis, which suggests that one way to gain decision speed is by investing in how quickly decision-makers can access or consider information.

Hypothesis 1: *Increasing the speed of information access leads to a faster decision, reducing the length of time taken for the organization to reach a decision.*

There are many ways in which firms invest in the speed of information access: adopting processes, practices, or norms for decision-making. One common way is through dashboards that display information in real-time, adopted by firms especially across tech, retail, and manufacturing (Berman and Israeli, 2022; Stein et al., 2025; Team, 2022)—which enable decision-makers to quickly access and attend to information without searching through or logging into access reports.

In many cases, the relationship between speed of information access and decision speed may be relatively straightforward: accessing information faster can allow a decision-maker to reach a decision more quickly by being able to consider the information sooner.

However, what is much less straightforward is how the increased speed of information access might affect the efficiency and quality of decisions – even beyond decision speed that in some cases can alone result in advantages relative to competitors. While much research on speed advantages implicitly assumes that accessing information more quickly will lead to the *same* decisions being made faster, we argue that increased speed may in fact change and improve how decisions are made, with resulting impacts on organizational performance beyond faster competitive moves.

First, we hypothesize that with faster information access, decision-makers can identify more relevant and targeted information, which may allow them to ultimately require less information

overall. Faster information access may enable decision-makers to be more adaptive in their information acquisition approach: they can make sequential decisions to learn from the early information they acquire to inform whether and which additional information they need, rather than seeking as much information as possible at once. Research on sequential and Bayesian experiments suggests that adaptive information acquisition can alter the course of experiments by improving their relevance and ensuring that efforts are concentrated on the most promising lines of inquiry (e.g., [Berry \(2004\)](#); [Mao and Bojinov \(2021\)](#); [Bojinov and Gupta \(2022\)](#)). For example, online contexts can generally enable faster information speed in collecting customer feedback compared to offline contexts, which has led many technology companies with online products to launch smaller early experiments that can provide targeted data to inform subsequent decisions on whether to build a new product feature.⁵ This more efficient decision-making process can then have impacts on organizational performance, by reducing the resources required and by extension the operational costs incurred. This leads us to our second hypothesis:

Hypothesis 2: *Increased speed in information access results in a more efficient decision-making process, leading to less, but more targeted information acquisition for decisions.*

Lastly, the speed of information access may enhance decision quality. This may manifest directly through more targeted information acquisition, or by enabling decision-makers to process and learn more effectively from information. Research in cognitive science suggests that the timing of stimuli affects what decision-makers notice and attend to at a time, due to cognitive constraints that restrict our span of memory and ability to process information ([Miller, 1956](#); [Lachman et al., 2015](#)). This also relates to the concept of bounded rationality, that a decision-maker does not have enough cognitive bandwidth to consider all relevant information – which has been seen as a key binding constraint to decision-making in the management literature ([Simon, 1991](#)). Similarly, having more considerations on one’s mind has been shown to decrease cognitive ability, as tasks compete for limited mental resources ([Mani et al., 2013](#)). These findings suggest that increasing the

⁵It is worth noting that there are many other differences between offline and online contexts, including the lower cost of acquiring information, which also may increase the total amount of information acquisition as well as its efficiency ([Koning et al., 2022](#)).

speed of information could enable decision-makers to receive information at a more optimal time when they hold the details of the decision in their memory without interruption in their attention—enabling better information processing. Instead of receiving information at a later time when other decisions have entered one’s mind and contextual details or possible hypotheses forgotten, accessing information more quickly may provide decision-makers with insights at the point of consideration, enabling them to make better inferences and ultimately better decisions. This leads to our third hypothesis.

***Hypothesis 3:** Increased speed in information access leads to improvements in the quality of decisions.*

In the following sections, we evaluate whether increasing the speed of information access not only leads to faster but also more efficient and higher quality decisions that improve organizational performance. Specifically, we investigate whether and how the speed of information access affects the information decision-makers choose to acquire and the resulting impacts on organizational performance.

3 Empirical context

We empirically examine how the speed of information access shapes the speed, efficiency, and quality of decisions in a healthcare context. We first outline the empirical requirements for studying this question and how our setting – a hospital emergency department – meets them. We then introduce the details of our setting and the technology intervention that increased the speed of information access.

3.1 Empirical setting

Studying the effects of information speed on decision-making imposes several empirical challenges. First, it requires access to a large number of complex, high-stakes decisions made by a diverse set of decision-makers. Because strategic decision-making often occurs under uncertainty, a large

sample of a common set of decisions is needed to evaluate whether decisions improve performance *in expectation*, as individual outcomes may not reveal underlying decision quality. Second, we must observe the full decision-making process, including what information is acquired, how it is used, how quickly decisions are made, and their performance consequences. This level of granularity is difficult to capture in most settings, as high-stakes decisions are rarely timestamped, documented in detail, or consistently tracked. Third, we need a substantial and plausibly exogenous change in the speed of information access to credibly isolate its effect from other factors, such as improved management practices or increased data availability.

To meet these requirements, we study a natural experiment in the Emergency Department of Fundación Valle del Lili (FVL), a leading teaching hospital in Cali, Colombia. Ranked 149th globally in Newsweek’s 2025 list of the World’s Best Hospitals ([Newsweek, 2025](#)), FVL operated 680 beds and 20 operating rooms in 2021, staffed by 724 physicians and serving nearly half a million outpatient visits and 36,000 hospital discharges annually. What makes this setting distinctive is that it provides real-time data on decisions made by hundreds of physicians, under high stakes, with detailed timestamps and measurable outcomes –features rarely available together. This allows us to track how decision-making changes when the speed of information access varies.

Our setting builds on a growing tradition in the strategic management literature that uses hospitals to study organizational decision-making. Prior work has leveraged healthcare settings to examine how organizations navigate uncertainty, allocate resources, and manage trade-offs between quality and efficiency. For example, [Douglas and Ryman \(2003\)](#) studied competitive advantage in U.S. hospitals, while [Nath and Sudharshan \(1994\)](#) analyze strategy coherence in acute care. Other studies have examined diversification in transplant centers ([Parker-Lue and Lieberman, 2020](#)), competitive technology adoption ([Sundaresan et al., 2023](#)), and cross-service synergies ([Huesch, 2013](#)). These precedents establish hospitals as insightful contexts for exploring strategic choices. Our contribution lies in examining how variation in the speed of information access affects organizational decisions and outcomes in a highly competitive healthcare environment. In doing so, we also build on a line of research that highlights how strategy is realized through the practical execution of

daily micro-level organizational decisions (Lee and Park, 2024; Jalonon et al., 2018; Healey and Hodgkinson, 2024). As Lee and Park (2024) underscore, this context illustrates how achieving an organization’s strategic goals hinges on frontline employees enacting those objectives in their operational decisions.

3.1.1 Data and decision structure

Between March and October of 2019 and 2022, the Emergency Department at FVL recorded over 64,000 patient encounters and more than 43,000 unique patients.

We focus on non-critical cases—patients who require urgent physician evaluation but are not in immediate danger. Upon arrival, patients are screened by administrative staff and a triage nurse. Those in critical condition (e.g., heart attacks, trauma) are sent directly to the resuscitation unit; patients with minor complaints are dismissed. The remaining patients are assigned to one of two Emergency Department wards based on insurance status and triage level: the regular ward (national insurance, triage levels 1–3) or the private insurance ward (private insurance, triage levels 1–5).⁶

Each patient typically interacts with two physicians. An initial consultation physician gathers history, symptoms, and test orders. The patient is then moved to a bed and cared for by an attending physician, who monitors progress, reviews test results, and decides whether to order more tests, continue observation, discharge the patient, or admit them to the hospital. We observe 387 attending physicians who made these final decisions.

We observe the full sequence of each decision: physician characteristics, patient condition (e.g., vitals, triage), the number and timing of lab tests ordered,⁷ the time spent, and the decision outcomes. Our outcome measures include length of stay (a proxy for decision speed), patient satisfaction, whether the patient was hospitalized, whether they returned to the ER within 30 days, and the total charges billed to the insurer.⁸

⁶Triage levels indicate clinical urgency, with level 1 being the most severe.

⁷The tests include blood counts, metabolic panels, urinalyses, and many others performed in-house.

⁸Charges include all billed items between ER admission and discharge: tests, drugs, consumables, and physician fees.

The outcomes we track map directly onto the dimensions of strategic positioning in healthcare, capturing more than just operational efficiency. Satisfaction and clinical outcomes are common indicators of quality differentiation, capturing both patient perceptions and health results. However, speed and responsiveness – such as length of stay – constitute a distinct and increasingly salient dimension of quality. For many patients, particularly in competitive settings, how quickly care is delivered is a key factor in their choice of provider. In contrast, costs reflect internal efficiency and throughput, and can drive a more value-based positioning. These trade-offs shape how hospitals strategically position themselves—whether by prioritizing speed and quality, minimizing costs, or attempting to balance all three.

Interviews with physicians and administrators confirmed that these metrics are strategically salient for the hospital. They highlighted that these attributes influence how they attract patients, negotiate with insurers, allocate resources, and monitor performance. Administrators noted that in particular, high-paying or well-insured patients often switch hospitals if dissatisfied, and emergency wards are acutely aware of these dynamics. These pressures influence how hospitals allocate resources, and highlights why faster access to information can matter strategically.

Increasing the speed of information access In 2022, FVL introduced a technology that improved physicians’ access to test results in one of the two wards. This change provided a rare opportunity to examine how faster access to information alters decision-making in this organizational context. Before the intervention, attending physicians had to manually log into a password-protected system and check each patient’s record to see whether test results were available.⁹ This task could not be delegated and required navigating multiple screens, leading to frequent delays.

An important feature of our setting is that the lab directly contacts the physician when a test

⁹Figure A.1 shows that observable variables of the ward like time, hour, day or patient characteristics are not good predictors of processing time (as the R2 is never large than 0.085). This implies that is very hard for doctors to predict the actual processing time of a test and check when it would be ready. On average, it takes about two hours to receive test results, and there are no statistically significant differences in processing times across wards or between the periods before and after the technology was introduced. Moreover, based on our conversations with physicians, test results are not reviewed so late that the information becomes obsolete—i.e., there is no evidence of depreciation of information in practice. While doctors do not check the results immediately upon release, they do so within a time frame that allows the information to remain clinically relevant and actionable.

result is critical, sharing the exact value. This is meant to prompt closer attention to the patient. Importantly, this process remained unchanged before and after the technology was introduced and was consistent across wards, so does not affect the interpretation of our empirical approach.

In June 2022, a new dashboard was installed in the private insurance ward. Developed in collaboration with a local software firm and the research team, the dashboard continuously displayed the status of all tests ordered for each patient in color-coded circles, without showing the test values.¹⁰ While physicians still needed to log in to view the numerical values, they could immediately see which results were ready.¹¹ The system offered passive display only, and no active alerts were sent.

The dashboard was installed in the third week of June 2022, with physician training completed by early July. We therefore define July 2022 as the start of the treatment period. This implementation created an environment where two nearly identical wards differed only in the speed at which attending physicians could access test results. This variation allows us to identify the causal effect of faster information access on decision speed, efficiency, and quality.¹²

4 Empirical strategy and data

In this section, we outline our empirical strategy and discuss the rationale behind the use of a triple-differences specification. We also explicitly outline our method for testing the parallel trends assumption and provide the main descriptive statistics of our data. Finally, we present arguments supporting the validity of our research design.

¹⁰Each time a new shift of doctors began, they were required to log into the dashboard; similarly, they logged out at the end of their shift. Logging in was necessary for the dashboard to receive and display information. The dashboard used color-coded circles to indicate the status of lab tests: white for pending, blue for validating, green for completed and normal, yellow for concerning, and red for critical. Importantly, the actual test values were not displayed – only these status indicators were shown.

¹¹Critical results were still communicated by phone, as before.

¹²There are many ways in which organizations can increase the speed of information access, which we remain agnostic on. In this particular case, reducing frictions to access it via this dashboard increased the speed of information access, but this may not always be the case. For example, removing logins for this hospital’s existing system without providing a notification for when test results are available may have reduced frictions to access it, but it may still not have meaningfully increased the speed at which physicians accessed the test results. Similarly, while increasing salience may generally increase the speed at which decision-makers access information, it may not always be the case. For example, if many items are highlighted or alarms fire frequently, users habituate, and salience loses its signaling power. In this paper, we focus on the impact of accelerating the time at which decision-makers access information, regardless of how it is implemented.

Differences-in-differences Our empirical strategy leverages a differences-in-differences (DiD) specification. Cases assigned to the private ward, where the technology was implemented from June 2022 onwards, represent the treatment group. Cases assigned to the regular ward, where the technology was never set up, represent the control group. Specifically, a baseline difference-in-differences model would estimate:

$$y_i = \beta(Private_{w(i)} \times Post_{t(i)}) + \alpha_{d(i)} + \theta_{in(i)} + \pi_{t(i)} + \gamma' \mathbf{X}_i + \epsilon_i \quad (1)$$

where y_i is an outcome (such as length of stay) of case i , $w(i)$ indexes the ward to which patient i is assigned, $d(i)$ indexes the physician allocated to patient i , and $t(i)$ indexes the exact hour (i.e. date and hour of day combination) in which patient i arrived. The model controls for physician $\alpha_{d(i)}$, insurance company $\theta_{in(i)}$ and hour $\pi_{t(i)}$ fixed effects, as well as patients' pre-determined characteristics $\mathbf{X}_i$ (patient age and gender, triage level, main diagnosis and patient vital signs on arrival to the ED).¹³ Our model includes insurance company and physician fixed effects, leveraging a key feature of the hospital: medical staff are regularly rotated across wards.¹⁴ This allows us to isolate the effect of the technology from physician-specific factors by comparing the same doctors across different settings.¹⁵ The parameter β captures the average differential outcome for cases assigned to the private insurance ward following the introduction of the technology.

Triple-differences strategy In the context of Emergency Departments, the exploitation of organizational changes to one hospital ward while using a different ward as a control group in a DiD strategy was pioneered by Chan (2016).¹⁶ The identification assumption in this type of strategy

¹³Instead of controlling for the ward to which the patient is assigned, we control more finely for the detailed insurance company of the patient as the insurance company fully determines the ward.

¹⁴Length of stay in an emergency department depends on a variety of factors, including the number of patients present, the laboratory's workload (i.e., the number of tests being processed), the availability of doctors, and the characteristics of the patient's condition. To account for the first three factors, we include detailed hour-by-hour fixed effects. To control for patient-level variation, we include the full set of observable characteristics measured at the time of the patient's arrival. In addition, we control for the identity of the attending physician.

¹⁵A small number of physicians appear only in one ward, typically because they spend limited time in the hospital. These cases do not contribute to the estimation, as the physician and insurance fixed effects are perfectly collinear. As a result, our identification relies on physicians who rotate across wards, strengthening our ability to compare their behavior in treatment and control conditions.

¹⁶We follow Chan (2016) in clustering the standard errors at the physician level.

is *not* that the expected outcomes across the two wards would have been similar in the absence of the treatment, an assumption that would clearly be violated in our setting. Instead, identification requires that the average outcomes across the two wards would have evolved similarly in the absence of the introduction of the technology.

A challenge in our setting is that, according to our discussions with FVL administrators and physicians, the introduction of the technology potentially coincided with seasonal changes in the composition of cases. Specifically, it may be that healthier patients (even after controlling for patient characteristics) may reach the private insurance ward in the summer months, relative to the winter months and to the regular ward.¹⁷ To alleviate this concern, our main estimating strategy is a triple-differences model comparing the months after June in the private insurance ward in 2022, relative to the regular ward and to 2019.¹⁸ Specifically, we pool the 2019 and 2022 March-October months together in the sample and use as the main independent variable of interest the triple interaction between $(Private_{w(i)} \times Post_{t(i)})$ and a year 2022 dummy. The model becomes:

$$y_i = \beta(Private_{w(i)} \times Post_{t(i)} \times 2022_{t(i)}) + \alpha_{d(i)} + \theta_{in(i)} + \pi_{t(i)} + \gamma'(\mathbf{X}_i \times 2022_{t(i)}) + \epsilon_i \quad (2)$$

In this triple-differences model, all the control variables (i.e., $\mathbf{X}_i$) are interacted with the year 2022 dummy to capture different evolution in the baseline characteristics of the patients.¹⁹

The main tables below report the results when the continuous (Length of Stay and Charges) and count measures (number of laboratory tests) are log-transformed. We do so to minimize the

¹⁷Alternatively, the number and composition of the medical staff present in the two wards may differ across the seasons, in a way that is not controlled by the physician fixed effects included in the regression. We have no anecdotal evidence that this is indeed the case, but it might be a potential concern.

¹⁸We chose the year 2019 as it is the last pre-COVID year. In 2020 and 2021 multiple changes to the internal organization of the ED (including the temporary elimination of the private insurance ward) make comparisons difficult. If instead, we choose 2018 instead of 2019, we obtain very similar estimates.

¹⁹A potential threat to the identification of our results is that the characteristics of the patients or physicians changed differentially across wards. Figure A.2 shows the coefficient estimates of our triple-difference model on different observable characteristics of the patients and physicians. The Figure shows that most of these interaction terms are not statistically different from 0, which provides evidence of the validity of our research design. Two variables show some negative and significant effects, gender (male) and age of the patient. This indicates that there is an increase in the fraction of younger men in the pool of patients in the private ward after the introduction of the technology. To deal with any potential changes in the composition of the pool of patients, we control explicitly for these variables below, so that all the results we present are robust to the potential change in the characteristics of the pool of patients.

impact that outliers can have on our results. All our results are consistent and qualitatively similar if instead of using the log-transformation, we use the Inverse Hyperbolic Sine (IHS) transformation.

Event study analysis The standard test of the identification assumption in the differences framework is the evaluation of potentially differential pre-trends. We evaluate these pre-trends using the following leads and lags model:

$$y_i = \sum_{j=-K \dots -1}^{1 \dots K} \beta_j (Private_{w(i)} \times Monthj_{t(i)}) + \alpha_{d(i)} + \theta_{in(i)} + \pi_{t(i)} + \gamma'(\mathbf{X}_i) + \epsilon_i \quad (3)$$

where $\hat{\beta}_{-K}, \dots, \hat{\beta}_{-1}$ capture the estimated effects of being assigned to the private insurance ward in the K months leading to the introduction of the technology, and $\hat{\beta}_1, \dots, \hat{\beta}_K$ capture the corresponding effects for the K months following the introduction in June 2022.²⁰

Descriptive statistics Our main analysis sample thus comprises eight months centered around the introduction of the technology in June 2022, and their equivalent in 2019. The 64,152 cases in our sample include 43,607 distinct patients cared for by 387 distinct physicians.

Table A.1 presents summary statistics for the main variables, organized into four panels: by period (before and after the dashboard introduction) and by ward type (private vs. non private). In the private ward, prior to the dashboard, the outcome variables exhibit a right-skewed distribution. For example, the average length of stay is 0.4 days, which is notably higher than the median of 0.17 days. This pattern of right-skewness is consistent across all four panels. In the same panel, the median patient undergoes 4.6 laboratory tests, the average logged episode charge is 13.6, 15% of patients are hospitalized, and 12% return to the emergency department within 30 days.

Comparing wards in the pre-dashboard period, patients in the private ward have lower average length of stay, fewer tests, lower charges, and lower hospitalization rates than those in the non

²⁰Note that our empirical specification is not affected by recent criticisms about DiD designs (de Chaisemartin and D’Haultfeuille, 2017; Callaway and Sant’Anna, 2021; Goodman-Bacon, 2021). First, treatment is not ‘fuzzy’ as defined in de Chaisemartin and D’Haultfeuille (2017) because no case is treated in the control group. Second, the treatment affects all the (treated) cases simultaneously and the private insurance ward remains treated for the remainder of the sample period. This rules out the concerns related to staggered treatment designs (Callaway and Sant’Anna, 2021; Goodman-Bacon, 2021).

private ward. Readmission rates are similar across wards. These patterns suggest that, even before the intervention, the private ward likely served healthier patients.²¹

Looking at variation over time, we see that the private ward shows little change in outcomes after the dashboard was introduced. In contrast, the non-private ward experiences noticeable increases in length of stay, number of tests, charges, and hospitalizations. According to hospital staff, these variables typically rise around July—the month immediately following the introduction of the technology—reflecting recurring annual cycles observed over multiple years. This seasonal pattern is precisely why our main econometric specification relies on a triple-difference design: we compare the post-June months in the private ward in 2022 not only to the non-private ward in the same year, but also to the same period in 2019, before the dashboard existed.

The absence of such increases in the private ward suggests that the introduction of the dashboard may have had a mitigating effect, and that this effect is visible even in the raw data. These raw patterns do not undermine our interpretation of the main results. Rather, the observed decreases in the private ward should be understood relative to the counterfactual of the non-private ward: without the dashboard, the private ward would almost certainly have shown similar increases.

— *Insert Table 1 Here* —

5 Information access speed and time to decision

In this section, we display the results of testing Hypothesis 1 that increasing the speed of information access leads to a faster decision. To do so, we measure the length of time to reach a decision using the patient’s length of stay in the Emergency ward, an important measure of organizational performance for Emergency departments in hospitals (Chan, 2016, 2018; Chan and Chen, 2022). While treatment times may vary across conditions, we do not expect this to affect our estimates provided that both wards observe similar variation in conditions.²² We also provide empirical evidence demonstrating that our research design seems to be valid, robust, and correctly identified.

²¹Table A.2 reports summary statistics for patient gender, age, and triage level. These characteristics remain relatively stable over time within each ward. Notably, in the pre-dashboard period, the private ward admits fewer triage 1 and 2 patients—the most severe cases—suggesting that patients in this ward are, on average, healthier. We control for these observables in all of our estimations.

²²While the two wards differ in triage levels, we control for them across our regression specifications.

Baseline results Table 1 displays the baseline results from running our triple-differences strategy according to equation 2. In this table and the ones that follow, standard errors are reported in parentheses. We start in Column 1 with a simplified model, which only controls for a post-June dummy and a private insurance ward dummy (both interacted with the year 2022 dummy). In Columns 2 and 3 we then sequentially introduce insurance status and hour-fixed effects, as well as (initial consultation) physician-fixed effects. From column 4 onwards, we add patient controls. Column 4 additionally winsorizes the top and bottom 1% of the length of stay distribution. While log-transforming this dependent variable should have alleviated the strong skewness of the length of stay distribution, winsorizing the top part of the distribution contributes to reassuring us that the baseline estimates are not disproportionately due to extreme positive values. We find that the introduction of patient controls and winsorizing the distribution has a strong effect on the estimate, decreasing it in absolute value from -0.35 to -0.16 . In column 5, we drop Triage 4 and 5 cases from the sample without winsorizing the distribution. Because these cases are only present in the private insurance ward, dropping them from the sample increases the homogeneity of the average case across the two wards, but substantially decreases our sample size. Finally, column 6 adds back the Triage 4 and 5 cases without winsorizing the distribution, and the treatment estimate remains broadly similar to column 5, and is more conservative than winsorization in column 4. Given this, we take this model as our baseline estimate.²³ The reduction of 13% represents a decrease of about 100 minutes with respect to the mean (the average episode lasts about 13 hours).

— Insert Table 2 Here —

These results support the anecdotal evidence we gathered. The physicians we interviewed believed they were making decisions slightly faster, but in a complex and stressful environment, they weren’t sure if their impressions were accurate. For instance, one physician noted, “... clearly, the screen helped reduce waiting time, since I no longer have to keep checking it repeatedly.”

To better understand the magnitude of these estimates and provide further validation to our triple-differences model, Table A.3 displays the baseline results from estimating Equation (1). We

²³We obtain similar results if we winsorize the distribution of the length of stay in this column.

find consistent results. The DiD model centered around July 2022 suggesting in Column 1 that the introduction of the technology is associated with a 25% decrease in the average length of stay. In Column 2, we repeat the estimation of (1) but in the ‘placebo’ year of 2019, in which no technology was introduced at any point. We find that the post-June period was associated with a decrease in length of stay of around 11%. While the decrease in the post-June months is much smaller in 2019 relative to 2022, it is still statistically significant. This suggests that there might be seasonal effects in the differential composition of patients arriving to the private insurance and regular wards. This confirms that a triple-differences model is more likely to be appropriate in our setting.

To better isolate the impact of the dashboard introduction, we further examine whether the reduction in length of stay is driven by critical or non-critical cases.²⁴ We expect the dashboard to have a stronger effect on non-critical cases, since in critical cases the laboratory directly contacts the physician, reducing the potential benefit of the dashboard. Table A.4 supports this hypothesis: the effect is negative for non-critical cases, and the difference in coefficients is statistically significant.

Leads and lags analysis We display in Figure 1 the estimates of (3). We do not find a significant differential effect of being assigned to the private insurance ward in the months leading to the introduction of the technology. In the first month period after this introduction, a discontinuous decrease in length of stay of around 35% is apparent in the figure. Overall, we interpret the evidence in Figure 1 as largely supportive of the main identification strategy in the paper.²⁵

Furthermore, Rambachan and Roth (2023) provides insight on how to conduct robust inference in designs that use difference-in-differences designs even if the assumption of parallel trends’ assumption may be violated. The idea is that we can estimate how large the post-treatment violations of the parallel trends have to be to make the design invalid. Figure A.3 conducts this robust inference and shows that the violation of the parallel trends assumption has to be very large (larger

²⁴An episode is classified as critical if at least one test result falls outside the defined criticality range. The number of such cases is low, as most test results lie within standard clinical thresholds.

²⁵As shown in Table A.1, the average length of stay in the raw data is lower in the private ward. However, Figure 1 suggests a possible positive difference in the pre-period, indicating a higher average length of stay in the pre-paid ward. This apparent discrepancy arises because the figure incorporates a rich set of controls, allowing for a more precise and comparable analysis across individuals.

than it could be, given Figure 1 and Table A.1) for our suggested effects to be invalid. Our results (here and for the other dependent variables) indicate that the estimated effect – a 25% to 35% reduction in length of stay following the introduction of the dashboard – is robust to small and moderate violations of the parallel trends assumption. The event study figures show no significant differences before the technology was introduced, supporting this assumption. Simulations further suggest that even with moderate deviations from parallel trends (e.g., 30% or 80%), the results remain statistically significant. However, when deviations become large (e.g., 130% or 180%), the estimates lose significance and are no longer robust. Importantly, our event study does not indicate any such large pre-treatment differences. The evidence from Figure 1 and A.3 provide further validity to our design, suggesting that the difference between the two emergency department wards remained similar before the intervention but not after.

— *Insert Figure 1 Here* —

In summary, these findings provide robust evidence in support of Hypothesis 1, showing that increasing the speed of information access leads to faster decisions. This effect may alone in some cases provide advantages for organizations, enabling them to make decisions before their competition, and suggests that one key lever through which organizations can realize speed advantages is through investing in the speed of information acquisition. In the next sections, we turn to the impact of faster information access on the efficiency and quality of decision-making.

6 Information access speed and information acquisition

In this section, we display results from testing Hypothesis 2 concerning the impact of information access speed on the efficiency of information acquisition, by examining the tests that physicians ordered – how they acquire information in this context – and the total costs to the health system.

6.1 Main results on information acquisition

We examine whether increasing the speed of information access led to changes in information acquisition choices by decision-makers, following Hypothesis 2. In Column 1 of Table 2, we replicate

our baseline specification to examine whether physicians changed the number of laboratory tests that they ordered. We find a precisely estimated negative coefficient: on average, the number of tests ordered decreased by 10% following the introduction of the technology. This finding suggests that increasing the speed of information led to more efficient information acquisition choices, decreasing the amount of overtesting as defined by the Chief Laboratory Officer at the hospital: “when a result is not relevant, it is lab test overutilization.”

— *Insert Table 3 Here* —

We confirm these findings in Figure 1, where, for the main variable of interest, we find that the number of tests ordered remained broadly unchanged in the months leading up to the introduction of the technology, and then decreased around 20% in August 2022, consistent with the equivalent figure for length of stay. The similarity in the evolution of these outcomes suggests that the decrease in the number of tests may have been a key mechanism enabling faster decisions.

This finding was also reflected in the qualitative interviews we conducted. Physicians felt that having faster access to the right information allowed them to rely on fewer tests overall. For example, one physician said, “This screen might be good for the hospital, because getting information faster makes me think twice about whether I really need to order the next test.”

Cases with and without tests We directly investigate if the increased speed in decisions arose from more efficient information acquisition, by exploring whether these effects were driven by cases that required ordering tests. If information access speed increases the speed of decisions through the mechanism of changing information acquisition choices, then we should find that cases that did not involve acquiring further information through laboratory tests (e.g., a sprained ankle) should not be affected strongly by the introduction of the technology. Indeed, we find evidence consistent with this hypothesis (see Table A.5 and Table A.6).²⁶

²⁶Table A.5 shows that the length of stay indeed decreases for cases in which tests were ordered, and not for cases in which they were not. Column 2 shows a slightly negative but statistically insignificant effect of the technology cases that do not involve ordering tests. However, the negative effect is precisely estimated and larger in magnitude when we focus on cases for which physicians order tests (column 3). In Table A.6, we find that the length of stay decreases by about 20%, the number of laboratory tests by 20%, the total charges by 17% and the probability of hospitalization by 4% for patients with a diagnosis that require tests *vis a vis* other patients with a diagnosis that do not require laboratory tests.

As an alternative approach, we use the main diagnosis of each episode to assess how frequently laboratory tests were ordered for that condition in the *past*. This approach relies on variation that is less directly influenced by the physician’s personal decision-making, in contrast to analyses based solely on whether tests were ordered—since that decision is made by the physician. Table A.7 presents the results of this analysis, which are fully consistent with those reported in the previous tables. We find that episodes involving diagnoses historically associated with frequent test ordering show the largest reductions in test use following the increase in information speed. The technology has a strong negative effect on test use for these diagnoses. In contrast, for diagnoses where lab tests are rarely ordered, the effect is close to zero.

To better understand how much this efficiency in information acquisition mediates the relationship between changes in information speed and length of stay, we conduct a mediation analysis following Heckman and Pinto (2015), the details of which we report in Section A.1. Our findings show that the mediator ratio is 48%, indicating that about half of the treatment effect stems from physicians’ ordering fewer tests.

Furthermore, we evaluate *how* information acquisition processes changed by exploring heterogeneity along two dimensions. First, we take advantage of the distinction between general tests – used across a wide range of conditions – and specialized tests, which are more targeted. Second, we consider the timing of test orders within a patient episode, distinguishing between those typically ordered early in the process and those more commonly ordered later.

Specialized and generic tests Generic tests are those that are frequently used across a wide range of conditions. For instance, Complete Blood Count (CBC) measures the levels of white and red blood cells as well as platelets. These tests are very generic as they are commonly ordered to assess overall health and detect conditions such as anaemia, infection, and leukaemia. Another example is the Urinalysis, which analyzes urine for various substances that can indicate different conditions, such as urinary tract infections, kidney disease, and diabetes. Other tests are more specific and targeted, such as the Angiotensin Converting Enzyme (ACE) test, which is used

to monitor patients with diagnosed sarcoidosis – a rare disease.

We thus define generic tests as those that were ordered in more than 50,000 cases (i.e., 1% of the tests) across all cases occurring between 2016-2022.²⁷ We find that physicians significantly reduced orders of generic tests, focusing more on targeted tests (Columns 2 and 3 in Table 3) – consistent with our hypothesis that information access speed led to more efficient information acquisition.

Timing of testing We also find that the largest reduction in the number of tests stemmed from the end of the decision, suggesting that earlier tests may have been effective in finding a promising path to diagnosis – also consistent with the hypothesis that information access speed enabled greater efficiency in information acquisition. Within the sample of cases where tests were ordered, we cut the length of stay into quartiles for each case. Columns 4-7 in Table 3 show that the reductions in the number of tests ordered stem mostly from the last quartile. This reinforces the idea that physicians were able to make more efficient information acquisition choices earlier on, decreasing the need to order additional tests at the end of their decision period.

6.2 Effect on Costs

We then explore whether these patterns of more efficient information acquisition led to lower direct costs billed to the health systems. Column 2 of Table 2 examines an outcome variable that indicates the total charges billed to the insurer associated with diagnosing a specific case. We find that this total charges variable decreased by around 25% following the introduction of the technology. Given that the average charge is about 1.4K US dollars, this represents a decrease of about 365 US dollars, approximately 1.5 times the local monthly minimum wage in 2022. These results are robust to winsorizing the charge variable. For instance, when we winsorize at 95% or 90%, the coefficient is still negative and significant at 1%, estimated at 24% and 22%, respectively. It is interesting to note that the effect of the decrease in the number of tests is far below the remarkable 25% decrease in charges from Table 2, suggesting that physicians eliminated relatively more expensive tests.

²⁷We find similar results when we cut the data in alternative ways.

Figure 1 provides supporting evidence for the parallel trends assumption with respect to costs. Specifically, it shows that, prior to the intervention, trends in patient charges were similar across the two wards. This suggests that any post-intervention differences in charges can more plausibly be attributed to the intervention itself, rather than to pre-existing differences in trends.

Overall, we find strong support for Hypothesis 2 that higher information access speed increases efficiency in information acquisition. Faster access to information reduced the number of tests ordered, especially by decreasing generic rather than specialized tests later in the medical episode. As a result, we also observe a corresponding reduction in total direct costs for the health system.

7 Information speed and organizational outcomes

In this section, we empirically investigate Hypothesis 3 on how the speed of information access affects organizational outcomes, focusing on the quality of care and patient satisfaction.

Hospitalization and Reincidence In Table 4, we examine the impact of increasing information speed on indicators of organizational performance. In Column 1 of Panel A, we find a large effect on the quality of care, decreasing the likelihood that the patient is hospitalized, rather than being discharged home. The decrease in this likelihood is around 7 percentage points, which represents around 27% of the unconditional likelihood. To the extent that admission to the hospital represents an acknowledgment that the patient has not improved sufficiently during her stay at the Emergency ward, we can conclude that increasing the speed of information access through the introduction of the technology improved patient outcomes.²⁸ In the context of overcrowded emergency departments, a decrease in hospitalization also typically implies less patient boarding.²⁹ Second, we find in Column 2 of Table 4 Panel A that this decrease in hospitalization happened without increasing the likelihood that the patient would return to the Emergency Department in

²⁸Conditional on two patients with the same initial condition (i.e., same initial vital signs, triage, entry time, physician), hospitalization is likely to reflect a worse treatment for the condition than being discharged home since it means that the patient was not able to recover sufficiently.

²⁹Boarding is keeping patients admitted to the Emergency Department for long periods due to inadequate capacity of inpatient wards. Boarding has been associated with negative effects on mortality and morbidity (Sartini et al., 2022).

the next month. The estimate is close to zero and is statistically insignificant, suggesting a safe discharge and thus an overall improvement in the quality of care.

— Insert Table 4 Here —

The leads and lags figures for these outcome variables can be found in Figure 1. For all outcomes, the graph shows no significant differences between the private insurance ward and the other ward in the months leading up to the introduction of the technology, providing further support for our findings. Moreover, Figure A.3 shows that the potential for violation of the parallel trend assumptions that would invalidate the research approach is much larger than they are according to Figure 1, providing additional support to the validity of our econometric design.

Patient satisfaction We additionally find that the introduction of the technology improved patient satisfaction with the care provided in the private insurance ward, relative to the regular ward (Table 4 Panel B). This may have been driven either by the decreased time taken to resolve their case or by the increased quality of care.

Our measures of patient satisfaction are based on a survey sent to a randomly-selected subset of patients. We focus on three responses to the survey: (1) whether the patient thought that the medical staff displayed the right attitude in the provision of care, (2) whether the patient thought that the physician complied with good medical practices, and (3) whether the patient believed that the physician was good at answering questions and addressing potential concerns.³⁰ Unfortunately, we only have these survey results for the year 2022 and for a sample size of fewer than two thousand observations. As a result, the specification for these variables is a DiD model that follows equation 1 (with coarser time effects), and results are noisier.

Panel B of Table 4 displays the results of these estimations. We find that the introduction of the technology had positive effects on patient satisfaction for all three measures. In terms of magnitude, the improvement on the attitude question is .2, which represents 37% of the .54 sample standard deviation. The increase in compliance and reported willingness to answer questions are slightly

³⁰Specifically, the questions in Spanish are ¿Cómo califica la atención médica? En terminos de: (1) Actitud de Servicio, (2) Cumplimiento en la cita, (3) Información y respuesta a sus inquietudes.

smaller at around 26% and 31%, respectively. We also use as a dependent variable the average of the three questions and find qualitatively similar results. The effects are large in magnitude, which illustrates how faster and improved decisions resulting from increasing information speed can have a meaningful impact on the way that patients feel about the hospital service.

In Figure A.4, we plot the estimated leads and lags from equation (1) of patient satisfaction. We find that patient satisfaction is not trending in any direction before July 2022, after which there is a meaningful improvement in all three measures as well as the average. Despite some noise in the estimates, Figures A.4 and A.3 (Honest DiD for the average of the three questions) suggest that technologies altering information speed can impact patient satisfaction.

To explore what might drive these improvements in quality of care and patient satisfaction, we explore the cognitive channel raised in Section 2, that the speed of information access may enhance decision quality by enabling decision-makers to process and learn more effectively from the information. These analyses are meant to be exploratory in nature, rather than providing indicative evidence for this mechanism.

To do so, we investigate two dimensions of heterogeneity: whether the ward was busy, and whether the diagnosis was common or uncommon. If information access speed affects decision quality by enabling decision-makers to better cognitively process the information, we would expect that faster information access should have a stronger effect when cognitive demands are higher. This could be the case during busy times, when physicians experience higher cognitive load due to having to switch between patients, which may reduce their ability to process and learn from information. It could also be the case for less common diagnoses, where lack of experience or familiarity requires physicians to think harder and spend more time reasoning through potential hypotheses and actions.

Ward busyness We find that the introduction of the technology indeed had a larger effect during busy hours (Table A.10). For each hour-ward combination, we construct the ratio of patients per physician and label the ward as busy if the ratio is larger than 2, which also represents the 75th

percentile of the distribution of patients per physician. Our interpretation of a busy ward is that physicians had more pressure to switch from one patient to another patient. When physicians need to switch across patients, it is likely more difficult to cognitively hold all details of each patient – whereas during slow times, physicians are tending to a single patient and it can be easy to remember and hold all details of that patient, regardless of the amount of time that has passed for processing the result. For all of our outcome variables, the magnitude of the treatment estimate during busy hours is nearly always more than double the estimate during non-busy hours. This provides suggestive, explorative evidence that receiving information in real-time may have enabled them to make better decisions by enabling them to better cognitively process the information when operating under constraints. It is also possible that the technology has a larger effect during busy times because it is precisely in these times that physicians may especially have long lags in checking test results, which can reduce the length of stay. However, this does not fully explain why we would observe larger improvements in other organizational outcomes.

Common and uncommon diagnoses We also examine whether the impact of the technology differed depending on how common the patient’s case was – which determines the amount of cognitive effort needed to make inferences and learn from the information acquired. If increasing information access speed helps decision-makers better process the information, we would expect that the effects should stem mainly from uncommon cases, which require physicians to think carefully through potential hypotheses and diagnoses, compared to more common cases where the diagnoses and necessary tests should be more routine and straightforward.

To investigate, we take the distribution of the universe of diagnoses of patients admitted to the hospital during the period 2016-2022. We label a diagnosis as uncommon if it belongs to the lowest 20th percentile of the distribution, and also examine robustness when cutting at the median. This distribution is heavily concentrated as a few diagnoses represent a large percentage of the population. Table [A.10](#) shows that the effects of the technology on the length of stay mainly occur in uncommon diagnoses when physicians must think through their information acquisition choices,

providing suggestive exploratory evidence.

Taken together, the results in this subsection show that increasing the speed of information access is associated with improved indicators of organizational performance beyond decision speed: higher quality of care and higher patient satisfaction. We also find suggestive evidence that this may be driven by a cognitive channel, where data velocity enables decision-makers to process and learn from the information more effectively.

8 Alternative Explanations and Other Results

In this section, we consider three possibilities to help interpret our results. First, doctors might be changing their behavior due to peer pressure from the new technology. Second, the quicker decisions—reflected in shorter patient stays—might be due to the lab processing test results faster for the private ward. Finally, we examine whether the impact of the change in information speed on various outcome variables may be driven by heterogeneity in physician characteristics.

Peer effects A potential alternative explanation is that the technology, by making test result statuses visible, could motivate physicians through peer effects—by increasing performance pressure when others are present. To explore, we compare treatment effects in shifts with multiple physicians present versus those with a single physician on duty. If peer effects were a primary driver, we would expect stronger effects when more than one physician is working. However, as shown in Table A.8, treatment effects are stronger when only one physician is present. This suggests that peer effects in the form of performance pressure are unlikely to be the main mechanism.³¹

Reaction of the laboratory test unit We investigate whether our findings can be attributed to expedited laboratory test processing for patients in the treated private ward. Our partner orga-

³¹Instead, based on interviews with physicians, we interpret this pattern as reflecting the collaborative nature of this setting: when two physicians are present, they often share responsibilities and rely on each other to monitor lab results. For example, when a physician receives a call from the lab about a critical result, they may also inquire about results for other physicians’ patients. As a result, the presence of a second physician can diffuse individual responsibility for checking the dashboard. In contrast, when working alone, physicians must monitor all results themselves, leading to greater engagement with the dashboard and larger observed effects.

nization maintains that this was not a contributing factor. To verify, we conduct triple-difference estimations using the duration of test processing as the dependent variable. The results, presented in Table A.9, indicate that the laboratory did not accelerate the processing of tests ordered for the private ward following the implementation of the technology *vis a vis* the processing of tests ordered for the non-private ward. This suggests that the observed effect on our measure of decision speed, length of stay, is unlikely to be driven by changes in laboratory test processing times.

Heterogeneity by Physician Characteristics We further explore whether physician heterogeneity might drive our results by exploring effects by their gender and age. Figure A.5 shows the estimated effects for our key outcomes, broken down by gender and experience. We do not find any statistically significant evidence that male doctors (compared to female doctors) or younger (vs. older) doctors were more affected by the change in how quickly information was delivered.³²

9 Discussion and conclusion

In this paper, we explore the impact of information speed on organizational decisions. We find that beyond speeding up a decision, increasing the speed of information access substantially improves indicators of organizational performance – improving the charges billed and quality of care in the ED of a leading hospital. Furthermore, we find that the speed of information access changes information acquisition choices, leading physicians to order fewer and more targeted tests earlier on, decreasing the need to order additional tests at the end of their decision period.

One notable aspect of our intervention is that decision-makers may not recognize the benefits of a new technology, even when those benefits accrue to the organization. Interviews with physicians revealed uncertainty about the positive impact of the technology. By the time the effects were analyzed, identified, and communicated to hospital staff, many physicians had already stopped logging into the system to view the dashboard. This aligns with evidence from other organizations

³²Doctors were classified as “Young” or “Old” based on their average age between 2019 and 2022. Those with an average age above the median were labeled “Old,” and those below the median, “Young.”

– particularly in healthcare – showing that decision-makers often underestimate the value of new technologies and resist their adoption (Kassirer, 2000; Benson, 2002; Lapointe and Rivard, 2006; Lin et al., 2012; Boeldt et al., 2015; Gawande, 2018). A key managerial implication is that the impacts of technological changes should be measured and communicated promptly to build organizational support and enable informed decisions about adoption and investment.

Of course, our study has several limitations. First, we are unable to observe all aspects of physician behavior – for example, the amount of time spent with each patient or the exact time at which physicians access test results, which limits our ability to more precisely test some of our hypotheses. We view both of these gaps as promising areas for future research that could help refine and extend our findings. Second, while our partner organization is broadly representative of emergency departments in both developed and developing settings, our analysis focuses on a single organization, which may limit the generalizability of our results. We believe that our results are more generalizable to similar settings as ours where the speed of decisions matter and earlier information acquisition choices can significantly impact later choices and decision outcomes.³³ We hope that future work can examine whether our findings hold in other contexts and help to identify the key boundary conditions that shape physician behavior in response to information technologies.

More broadly, these findings suggest that we need to think more deeply about how organizations acquire and learn from information in the digital age. While there has been much focus on the acquisition or collection of information, one implication of our findings is that organizations may not need to accelerate information collection, but rather invest in tools that reduce the speed of accessing and recognizing the information they already have.

Furthermore, our findings highlight the strategic importance of investing in systems that give frontline decision-makers faster access to critical information, especially in healthcare settings. Even small reductions in the time between when a diagnostic test is processed and when a physician sees

³³Highlighting the importance of speed in this competitive context, the Chief Laboratory Officer of the hospital remarked: “What led the hospital to undertake this pilot was the need to be more efficient in the ER, which in Colombia is an activity that generates economic losses. The most important variable is time. EDs frequently have over-occupancy of 120 to 150% or more.” (Chief Laboratory Officer)

the results can lead to better outcomes – not only by shortening patients’ length of stay but also by helping physicians make better, more targeted decisions. For hospital administrators and operations managers, this means that speeding up access to information should not be seen only as an IT upgrade or process improvement. Instead, it should be viewed as a key strategic tool to improve both efficiency and quality of care.

More broadly, in similarly fast-paced environments where decisions must be made quickly and with uncertainty, investing in information speed may be a powerful and scalable way for organizations to learn and adapt – especially in situations where judgment under time pressure is essential. If a marginal improvement in information speed can result in substantial speed-based advantages and enable key decision-makers to improve their choices, this suggests that how organizations build in mechanisms to learn from data may increasingly become a key source of competitive advantage.

References

- Adner, R. and Levinthal, D. A. (2004). What is not a real option: Considering boundaries for the application of real options to business strategy. *Academy of Management Review*, 29(1):74–85.
- Adner, R., Puranam, P., and Zhu, F. (2019). What is different about digital strategy? from quantitative to qualitative change. *Strategy Science*, 4(4):253–261.
- Agarwal, R., Bacco, F., Camuffo, A., Coali, A., Gambardella, A., Msangi, H., Sonka, S. T., Temu, A., Waized, B., and Wormald, A. (2023). Does a theory-of-value add value? evidence from a randomized control trial with tanzanian entrepreneurs. *Working Paper*.
- Agrawal, A., Camuffo, A., Gambardella, A., Gans, J., Scott, E., and Stern, S. (2024). Bayesian entrepreneurship. *White Paper*.
- Agrawal, A., Gans, J. S., and Goldfarb, A. (2019). Artificial intelligence: the ambiguous labor market impact of automating prediction. *Journal of Economic Perspectives*, 33(2):31–50.
- Allen, R. T. and McDonald, R. M. (2025). Methodological pluralism and innovation in data-driven organizations. *Administrative Science Quarterly*, page 00018392251313737.
- Barends, E. and Rousseau, D. M. (2018). *Evidence-based management: How to use evidence to make better organizational decisions*. Kogan Page Publishers.
- Baum, R. and Wally, S. (2003). Strategic decision speed and firm performance. *Strategic Management Journal*, 24(11):1107–1129.
- Benson, T. (2002). Why general practitioners use computers and hospital doctors do not—part 1: incentives. *Bmj*, 325(7372):1086–1089.
- Berman, R. and Israeli, A. (2022). The value of descriptive analytics: Evidence from online retailers. *Marketing Science*, 41(6):1074–1096.
- Berry, D. A. (2004). Bayesian statistics and the efficiency and ethics of clinical trials.
- Bharadwaj, A., El Sawy, O. A., Pavlou, P. A., and Venkatraman, N. (2013). Digital business strategy: Toward a next generation of insights. *MIS Quarterly*, 37(2):471–482.
- Blank, S. (2013). Why the lean startup changes everything. *Harvard Business Review*.
- Bloom, N., Garicano, L., Sadun, R., and Van Reenen, J. (2014). The Distinct Effects of Information Technology and Communication Technology on Firm Organization. *Management Science*, 60(12):2859–2885.
- Boeldt, D. L., Wineinger, N. E., Waalen, J., Gollamudi, S., Grossberg, A., Steinhubl, S. R., McCollister-Slipp, A., Rogers, M. A., Silvers, C., and Topol, E. J. (2015). How consumers and physicians view new medical technology: comparative survey. *Journal of medical Internet research*, 17(9):e215.
- Bojinov, I. and Gupta, S. (2022). Online experimentation: Benefits, operational and methodological challenges, and scaling guide.
- Brogaard, J., Hendershott, T., and Riordan, R. (2014). High frequency trading and price discovery. *The Review of Financial Studies*, 27(8):2267–2306.
- Brynjolfsson, E. and McElheran, K. (2016). The rapid adoption of data-driven decision-making. *American Economic Review*, 106(5):133–39.

- Brynjolfsson, E. and McElheran, K. (2019). Data in action: data-driven decision making and predictive analytics in us manufacturing. *Rotman School of Management Working Paper*, (3422397).
- Callaway, B. and Sant'Anna, P. H. (2021). Difference-in-Differences with Multiple Time Periods. *Journal of Econometrics*, 225(2):200–230.
- Camuffo, A., Cordova, A., Gambardella, A., and Spina, C. (2020). A scientific approach to entrepreneurial decision making: Evidence from a randomized control trial. *Management Science*, 66(2):564–586.
- Camuffo, A., Gambardella, A., and Pignataro, A. (2023). Theory-driven strategic management decisions. Technical report, CEPR Discussion Papers.
- Chan, D. C. (2016). Teamwork and Moral Hazard: Evidence from the Emergency Department. *Journal of Political Economy*, 124(3):734–770.
- Chan, D. C. (2018). The Efficiency of Slacking off: Evidence From the Emergency Department. *Econometrica*, 86(3):997–1030.
- Chan, D. C. and Chen, Y. (2022). The Productivity of Professions: Evidence from the Emergency Department. Working Paper 30608, National Bureau of Economic Research.
- Chen, H., Chiang, R. H., and Storey, V. C. (2012). Business intelligence and analytics: From big data to big impact. *MIS Quarterly*, 36(4):1165–1188.
- Davenport, T. H. and Patil, D. J. (2012). Data scientist: The sexiest job of the 21st century. *Harvard Business Review*, 90(10):70–76.
- de Chaisemartin, C. and D'Haultfeuille, X. (2017). Fuzzy Differences-in-Differences. *The Review of Economic Studies*, 85(2):999–1028.
- Deck, C. and Jahedi, S. (2015). The effect of cognitive load on economic decision making: A survey and new experiments. *European Economic Review*, 78:97–119.
- Douglas, T. J. and Ryman, J. A. (2003). Understanding competitive advantage in the general hospital industry: Evaluating strategic competencies. *Strategic Management Journal*, 24(4):333–347.
- Eisenhardt, K. M. (1989). Making fast strategic decisions in high-velocity environments. *Academy of Management Journal*, 32(3):543–576.
- Eisenmann, T. R., Ries, E., and Dillard, S. (2012). *Hypothesis-driven entrepreneurship: The lean startup*. SSRN.
- Forbes, D. P. (2005). Managerial determinants of decision speed in new ventures. *Strategic Management Journal*, 26(4):355–366.
- Galdon-Sanchez, J. E., Gil, R., and Uriz Uharte, G. (2024). The value of information in competitive markets: The impact of big data on small and medium enterprises. *Forthcoming, Journal of Political Economy*.
- Gans, J. S. (2023). Experimental choice and disruptive technologies. *Management Science*, 69(11):7044–7058.
- Gans, J. S., Stern, S., and Wu, J. (2019). Foundations of entrepreneurial strategy. *Strategic Management Journal*, 40(5):736–756.
- Garduno, C. (2022). How big data is helping advertisers solve problems. *Forbes*.
- Garud, R., Jain, S., and Phelps, C. (1998). A tale of two browsers. *New York University Teaching Case*.
- Gawande, A. (2018). Why doctors hate their computers. *The New Yorker*, 12.
- Goldfarb, A. and Tucker, C. (2019). Digital economics. *Journal of Economic Literature*, 57(1):3–43.
- Goodman-Bacon, A. (2021). Difference-in-Differences with Variation in Treatment Timing. *Journal of Econometrics*, 225(2):254–277.
- Hawk, A., Pacheco-De-Almeida, G., and Yeung, B. (2013). Fast-mover advantages: Speed capabilities and entry into the emerging submarket of atlantic basin lng. *Strategic Management Journal*, 34(13):1531–1550.
- Hawser, A. (2022). Banks' eyes in the sky. *The Banker*.
- Healey, M. P. and Hodgkinson, G. P. (2024). Overcoming strategic persistence: Effects of multiple scenario analysis on strategic reorientation. *Strategic Management Journal*, 45(8):1423–1445.
- Heckman, J. J. and Pinto, R. (2015). Econometric Mediation Analyses: Identifying the Sources of Treatment Effects from Experimentally Estimated Production Technologies with Unmeasured and Mismeasured Inputs. *Econometric Reviews*, 34(1-2):6–31.
- Helfat, C. E., Finkelstein, S., Mitchell, W., Peteraf, M., Singh, H., Teece, D., and Winter, S. G. (2009). *Dynamic capabilities: Understanding strategic change in organizations*. John Wiley & Sons.
- Hoelzemann, J., Manso, G., Nagaraj, A., and Tranchero, M. (2024). The streetlight effect in data-driven exploration. Technical report, National Bureau of Economic Research.
- Huesch, M. D. (2013). Are there always synergies between productive and allocative efficiency? evidence from hospitals. *Strategic Management Journal*, 34(8):954–970.
- Jalonen, K., Schildt, H., and Vaara, E. (2018). Strategic concepts as micro-level tools in strategic sensemaking. *Strategic Management Journal*, 39(10):2794–2826.
- Kassirer, J. P. (2000). Patients, physicians, and the internet: Coming generations of doctors are ready to embrace new technology, but few incentives now exist to encourage their older peers to do likewise. *Health Affairs*, 19(6):115–123.
- Kerr, W. R., Nanda, R., and Rhodes-Kropf, M. (2014). Entrepreneurship as experimentation. *Journal of Economic Perspectives*, 28(3):25–48.
- Kim, H. (2024). The value of competitor information: Evidence from a field experiment. Technical Report 4.
- Kim, H. (2025). Learning from Data in Entrepreneurial Experimentation. Bayesian Entrepreneurship. MIT Press.
- Kim, H., Glaeser, E. L., Hillis, A., Kominers, S. D., and Luca, M. (2024). Decision authority and the returns to algorithms. *Strategic Management Journal*, 45(4):619–648.

- Koning, R., Hasan, S., and Chatterji, A. (2022). Experimentation and start-up performance: Evidence from a/b testing. *Management Science*, 68(9):6434–6453.
- Lachman, R., Lachman, J. L., and Butterfield, E. C. (2015). *Cognitive psychology and information processing: An introduction*. Psychology Press.
- Langley, A. (1995). Between ‘paralysis by analysis’ and ‘extinction by instinct’. *MIT Sloan Management Review*, 36(3):63.
- Lapointe, L. and Rivard, S. (2006). Getting physicians to accept new information technology: insights from case studies. *Cmaj*, 174(11):1573–1578.
- Leatherbee, M. and Katila, R. (2020). The lean startup method: Early-stage teams and hypothesis-based probing of business ideas. *Strategic Entrepreneurship Journal*, 14(4):570–593.
- Lee, S. and Park, J. (2024). Giving up learning from failures? an examination of learning from one’s own failures in the context of heart surgeons. *Strategic Management Journal*, 45(10):2063–2094.
- Levinthal, D. A. (2017). Mendel in the c-suite: Design and the evolution of strategies. *Strategy Science*, 2(4):282–287.
- Lin, C., Lin, I.-C., and Roan, J. (2012). Barriers to physicians’ adoption of healthcare information technology: an empirical study on multiple hospitals. *Journal of medical systems*, 36:1965–1977.
- Makadok, R. (1998). Can first-mover and early-mover advantages be sustained in an industry with low barriers to entry/imitation? *Strategic Management Journal*, 19(7):683–696.
- Mani, A., Mullainathan, S., Shafr, E., and Zhao, J. (2013). Poverty impedes cognitive function. *science*, 341(6149):976–980.
- Mao, J. and Bojinov, I. (2021). Quantifying the value of iterative experimentation. *arXiv preprint arXiv:2111.02334*.
- McAfee, A., Brynjolfsson, E., Davenport, T. H., Patil, D., and Barton, D. (2012). Big data: the management revolution. *Harvard Business Review*, 90(10):60–68.
- Miller, G. A. (1956). The magical number seven, plus or minus two: Some limits on our capacity for processing information. *Psychological Review*, 63(2):81.
- Mosakowski, E. (1997). Strategy making under causal ambiguity: Conceptual issues and empirical evidence. *Organization Science*, 8(4):414–442.
- Nagaraj, A. (2022). The private impact of public data: Landsat satellite maps increased gold discoveries and encouraged entry. *Management Science*, 68(1):564–582.
- Nath, D. and Sudharshan, D. (1994). Measuring strategy coherence through patterns of strategic choices. *Strategic Management Journal*, 15(1):43–61.
- Newsweek (2025). World’s best hospitals 2025. Available at: <https://rankings.newsweek.com/worlds-best-hospitals-2025>.
- Pacheco-de Almeida, G., Hawk, A., and Yeung, B. (2015). The right speed and its value. *Strategic Management Journal*, 36(2):159–176.
- Parker-Lue, S. and Lieberman, M. B. (2020). The impact of diversification on task performance: Evidence from kidney transplant centers. *Strategic Management Journal*, 41(7):1280–1307.
- Perlow, L. A., Okhuysen, G. A., and Repenning, N. P. (2002). The speed trap: Exploring the relationship between decision making and temporal context. *Academy of Management Journal*, 45(5):931–955.
- Pfeffer, J. and Sutton, R. I. (2000). *The knowing-doing gap: How smart companies turn knowledge into action*. Harvard Business Press.
- Pfeffer, J. and Sutton, R. I. (2006). Evidence-based management. *Harvard Business Review*, 84(1):62.
- Rambachan, A. and Roth, J. (2023). A more credible approach to parallel trends. *Review of Economic Studies*, 90(5):2555–2591.
- Ries, E. (2011). *The lean startup: How today’s entrepreneurs use continuous innovation to create radically successful businesses*. Crown Currency.
- Sartini, M., Carbone, A., Demartini, A., Giribone, L., Oliva, M., Spagnolo, A. M., Cremonesi, P., Canale, F., and Cristina, M. L. (2022). Overcrowding in emergency department: causes, consequences, and solutions—a narrative review. In *Healthcare*, volume 10, page 1625. MDPI.
- Shi, W., Cao, J., Zhang, Q., Li, Y., and Xu, L. (2016). Edge computing: Vision and challenges. *IEEE Internet of Things Journal*, 3(5):637–646.
- Simon, H. A. (1991). Bounded rationality and organizational learning. *Organization Science*, 2(1):125–134.
- Stalk, G. (1988). Time—the next source of competitive advantage. *Harvard Business Review*.
- Stalk Jr, G. and Hout, T. M. (1990). Competing against time. *Research-Technology Management*, 33(2):19–24.
- Stein, E., Robinson, K., Wolfer, A., Almeida, G., and Huang, W. (2025). Unlocking the next frontier of personalized marketing. <https://www.mckinsey.com/capabilities/growth-marketing-and-sales/our-insights/unlocking-the-next-frontier-of-personalized-marketing>. [Online].
- Sundaresan, S., Boysen, A., and Nerkar, A. (2023). Adopting dr. robot: Responses to competitor adoption in the market for surgical robots. *Strategic Management Journal*, 44(3):656–683.
- Team, U. (2022). Product analytics dashboard: Which metrics to track and how to create one. <https://userpilot.medium.com/product-analytics-dashboard-which-metrics-to-track-and-how-to-create-one-98ba745c5e85>. [Online].
- Teece, D. J., Pisano, G., and Shuen, A. (1997). Dynamic capabilities and strategic management. *Strategic Management Journal*, 18(7):509–533.

Figures and Tables

**TABLE 1: THE IMPACT OF INFORMATION SPEED
ON TIME TO DECISION**

Dependent Variable = Log Length of Stay						
	(1)	(2)	(3)	(4)	(5)	(6)
Post June X Private Ward X Year 2022	-.254 (.063)	-.383 (.089)	-.352 (.085)	-.163 (.049)	-.124 (.059)	-.133 (.051)
Post June X Private Ward	Yes	Yes	Yes	Yes	Yes	Yes
Patient Controls X Year 2022	No	No	No	Yes	Yes	Yes
Doctor Fixed Effects X Year 2022	No	No	Yes	Yes	Yes	Yes
Insurance Status Fixed Effects X Year 2022	No	Yes	Yes	Yes	Yes	Yes
Date X Hour Fixed Effects	No	Yes	Yes	Yes	Yes	Yes
Post June X Year 2022	Yes	No	No	No	No	No
Private Ward X Year 2022	Yes	No	No	No	No	No
Observations	63,025	61,750	61,612	62,618	40,830	61,600

Note: This Table displays estimates of regressions of a case's length of stay in the ED on the period during which the technology was introduced (i.e. after June), interacted with the ward in which it was introduced (i.e. private ward) and with the year 2022 dummy. The unit of observation is a case i arriving at the ED. The estimating equation in Column 4 is:

$$y_i = \beta(Private_{w(i)} \times Post_{t(i)} \times 2022_{t(i)}) + \alpha_{d(i)} + \theta_{w(i)} + \pi_{t(i)} + \gamma'(\mathbf{X}_i \times 2022_{t(i)}) + \epsilon_i$$

where w indexes the ward to which the patient is assigned, t indexes the exact hour (i.e. date/hour of day combination) in which the patient arrived and d indexes the physician to which the patient was assigned. The main independent variable of interest is the interaction between being assigned to the private ward, arriving in the months between July and October, and arriving in 2022. The model controls for insurance status (which subsumes the assigned ward), physician and hour-fixed effects, as well as patient controls (age, gender, dummies for the main diagnosis, as well as dummies for extreme vital signs markers upon admission). All the patient controls are interacted with the year 2022 dummy. In Column 1 we display the most streamlined triple-differences model, which only includes a post dummy and a private dummy as controls, interacted with each other and with the year 2022 dummy, this regression includes all triages in its estimation and length of stay is not winsorized. In Column 2 we include the insurance status and the hour fixed effects (which subsume the year 2022 and private ward dummies), this regression includes all triages in its estimation, and the length of stay is not winsorized. In Column 3 we add the physician fixed effects, this regression includes all triages in its estimation, and length of stay is not winsorized. In Column 4 we add the patient controls and the top and bottom 1% of the length of stay distribution is winsorized, this regression includes all triage in its estimation. In Column 5 we add the patient controls and the sample includes only triage levels 1–3 and the length of stay is not winsorized. Column 6 is the baseline model for the full sample, this regression includes all triages in its estimation, and length of stay is not winsorized. Standard errors are clustered at the physician level.

**TABLE 2: THE IMPACT OF INFORMATION SPEED ON
TESTS AND TOTAL CHARGES**

Dependent Variable =	(1) Log. Tests	(2) Total Cost
Post June X Private Ward X Year 2022	-.102 (.037)	-.255 (.059)
Patient Controls X Year 2022	Yes	Yes
Doctor Fixed Effects X Year 2022	Yes	Yes
Insurance Status Fixed Effects X Year 2022	Yes	Yes
Date X Hour Fixed Effects	Yes	Yes
Observations	62,650	62,546

Note: This table displays estimates of regressions of a case's different medical outcomes in the ED on the period during which the technology was introduced (i.e. after June), interacted with the ward in which it was introduced (i.e. private ward) and with the year 2022 dummy. The estimating equation is:

$$y_i = \beta(Private_{w(i)} \times Post_{t(i)} \times 2022_{t(i)}) + \alpha_{d(i)} + \theta_{w(i)} + \pi_{t(i)} + \gamma'(\mathbf{X}_i \times 2022_{t(i)}) + \epsilon_i$$

The model controls for insurance status (which subsumes the assigned ward), physician and hour-fixed effects, as well as patient controls (age, gender, dummies for the main diagnosis, as well as dummies for extreme vital signs markers upon admission). In Column 1, the dependent variable is the log. number of tests. In column 2, the dependent variable is the total charge (defined as the log of the amount in Colombian Pesos (COP) charged by the hospital to the payer for each episode). All the controls are interacted with the year 2022 dummy. Standard errors are clustered at the physician level.

TABLE 3: CHANGES IN THE INFORMATION ACQUISITION PROCESS

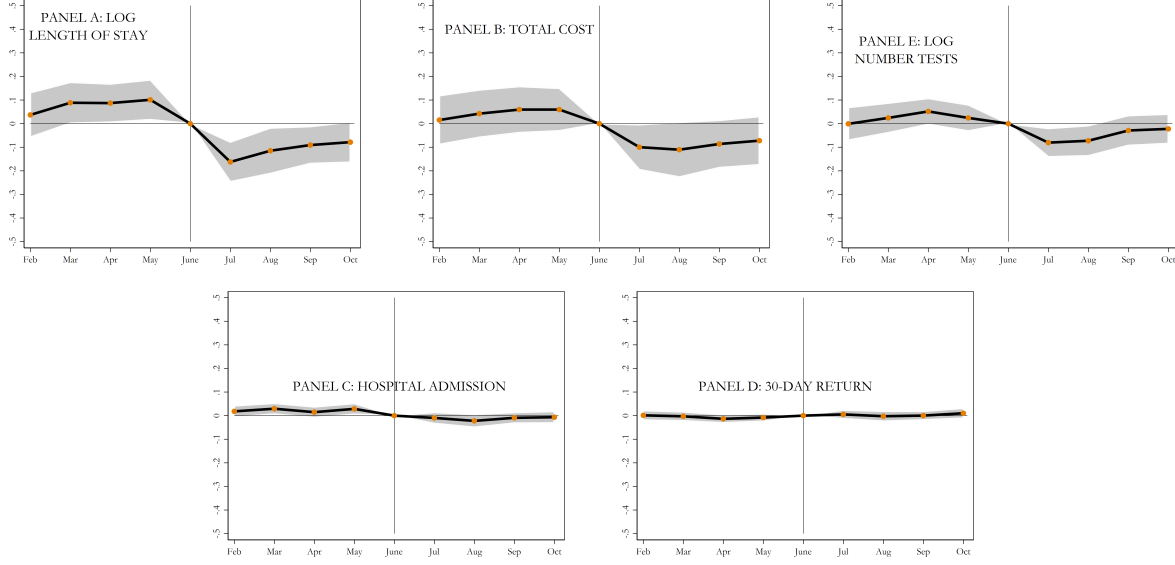
	Type of Tests		Percentiles			
	Targeted	Generic	0–25	25–50	50–75	75–100
Post June X Private Ward X Year 2022	-.042 (.032)	-.102 (.034)	-.021 (.034)	-.117 (.123)	-.001 (.326)	-2.335 (.53)
Post June X Private Ward	Yes	Yes	Yes	Yes	Yes	Yes
Patient Controls X Year 2022	Yes	Yes	Yes	Yes	Yes	Yes
Doctor Fixed Effects X Year 2022	Yes	Yes	Yes	Yes	Yes	Yes
Insurance Status Fixed Effects X Year 2022	Yes	Yes	Yes	Yes	Yes	Yes
Date X Hour Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	62,650	62,650	30,163	5,886	1,483	1,315

Note: This Table displays estimates of regressions of the log. of the number of tests on the period during which the technology was introduced (i.e. after June), interacted with the ward in which it was introduced (i.e. private ward) and with the year 2022 dummy. To determine whether the test is generic, we defined it as 1 (generic) if the test category appears in 50.000 (the median value) or more episodes and 0 otherwise. 50.000 episodes represent the 99% percentile of the distribution of laboratory tests. The last four columns show the effect of the interaction term on the log number of tests cutting the sample into 4 quartiles of the length of stay. For instance, the column 0-25 shows the effect of the change of the speed on information on the number of tests for the first quarter of the episode. The estimating equation is:

$$y_i = \beta(Private_{w(i)} \times Post_{t(i)} \times 2022_{t(i)}) + \alpha_{d(i)} + \theta_{w(i)} + \pi_{t(i)} + \gamma'(\mathbf{X}_i \times 2022_{t(i)}) + \epsilon_i$$

where w indexes the ward to which the patient is assigned, t indexes the exact hour (i.e. date/hour of day combination) in which the patient arrived and d indexes the physician to which the patient was assigned. The main independent variable of interest is the interaction between being assigned to the private ward, arriving in the months between July and October, and arriving in 2022. The model controls for insurance status (which subsumes the assigned ward), physician and hour fixed effects, as well as patient controls (age, gender, dummies for the main diagnosis, as well as dummies for extreme vital signs markers upon admission) interacted with the year 2022 dummy. Standard errors are clustered at the physician level.

FIGURE 1: LEADS AND LAGS EVIDENCE



This Figure displays dynamic estimates of regressions of a case's length of stay in the ED on the period during which the technology was introduced and interacted with the ward in which it was introduced (i.e. private ward). The unit of observation is a case i arriving at the ED. This figure displays the coefficients β_j from estimating:

$$y_i = \sum_{t=2}^{10} \beta_j (Private_{w(i)} \times Monthj_{t(i)}) + \alpha_{d(i)} + \theta_{w(i)} + \pi_{t(i)} + \gamma' \mathbf{X}_i + \epsilon_i$$

where w indexes the ward to which the patient is assigned, t indexes the exact hour (i.e. date/hour of day combination) in which the patient arrived, and d indexes the physician to which the patient was assigned. The model controls for insurance status (which subsumes the assigned ward), physician and hour-fixed effects, as well as patient controls (age, gender, and health markers upon admission). Standard errors are clustered at the physician level. To show longer dynamics we increase our sample from March to October (as the rest of the paper), to February to October.

**TABLE 4: THE IMPACT OF INFORMATION SPEED ON
OTHER ORGANIZATIONAL OUTCOMES**

PANEL A: EFFECTS ON QUALITY OF CARE

	(1)	(2)
Dependent Variable =	Hospitalization	30-Day Return
Post June X Private Ward X Year 2022	-.071 (.014)	.01 (.013)
Patient Controls X Year 2022	Yes	Yes
Doctor Fixed Effects X Year 2022	Yes	Yes
Insurance Status Fixed Effects X Year 2022	Yes	Yes
Date X Hour Fixed Effects	Yes	Yes
Observations	62,650	62,650

PANEL B: EFFECTS ON PATIENT SATISFACTION

	(1)	(2)	(3)	(4)
	Attitude	Compliance	Answering Questions	Average
Post June X Private Ward	.201 (.059)	.156 (.064)	.18 (.058)	.171 (.057)
Patient Controls	Yes	Yes	Yes	Yes
Doctor Fixed Effects	Yes	Yes	Yes	Yes
Insurance Status Fixed Effects	Yes	Yes	Yes	Yes
Month Fixed Effects	Yes	Yes	Yes	Yes
Mean Dep. Var.	3.68	3.6	3.65	3.62
SD Dep. Var.	.54	.59	.58	.54
Observations	1,445	1,233	1,443	1,232

Note: Panel A displays estimates of regressions of a case's different medical outcomes in the ED on the period during which the technology was introduced (i.e. after June), interacted with the ward in which it was introduced (i.e. private ward) and with the year 2022 dummy. The estimating equation is:

$$y_i = \beta(Private_{w(i)} \times Post_{t(i)} \times 2022_{t(i)}) + \alpha_{d(i)} + \theta_{w(i)} + \pi_{t(i)} + \gamma'(\mathbf{X}_i \times 2022_{t(i)}) + \epsilon_i$$

The model controls for insurance status (which subsumes the assigned ward), physician and hour-fixed effects, as well as patient controls (age, gender, dummies for the main diagnosis, as well as dummies for extreme vital signs markers upon admission). In Column 1, the dependent variable is a dummy if the patient is hospitalized. In Column 2, the dependent variable is a dummy that takes the value of 1 if the patient returns to the ED within a 30-days period. All the controls are interacted with the year 2022 dummy. Panel B displays estimates of regressions of patients' evaluations in the ED on the period during which the technology was introduced, and interacted with the ward in which it was introduced. The estimating equation is:

$$y_i = \beta(Private_{w(i)} \times Post_{t(i)}) + \alpha_{d(i)} + \theta_{w(i)} + \pi_{t(i)} + \gamma' \mathbf{X}_i + \epsilon_i$$

The main independent variable of interest is the interaction between being assigned to the private ward and arriving in the month of July or after. The model controls for insurance status (which subsumes the assigned ward), physician and month-fixed effects, as well as patient controls (age, gender, dummies for the main diagnosis, as well as dummies for extreme vital signs markers upon admission). For both panels where w indexes the ward to which the patient is assigned, t indexes the month in which the patient arrived, and d indexes the physician to which the patient was assigned. The unit of observation is a case i arriving to the ED. Standard errors are clustered at the physician level.

Online Appendix

A.1 Mediation analysis

While identifying the quantitative importance of different mechanisms is a notoriously difficult exercise, in this subsection we examine whether any share in the decrease in the length of stay may be independent from the effects on the number of tests. Firstly, we estimate equation (2) while controlling for the number of tests ordered. We display the results in Column 4 Table A.3. We find there that the baseline triple-differences coefficient decreases in magnitude, and becomes not statistically significant. That is, the decrease in the length of stay becomes much smaller if one holds the number of tests ordered constant. We interpret this result as additional supportive evidence on the information acquisition mechanism – that increasing information speed had an effect in organizational outcomes through the laboratory tests that physicians chose to order.

While enlightening, the evidence in Column 4 Table A.3 can at best only be regarded as suggestive, given that the regressions are controlling for explicitly endogenous variables. For a more systematic analysis, we follow Heckman and Pinto (2015) in quantifying the relative importance of our mediating variable in the estimated decrease in length of stay. Heckman and Pinto (2015) consider an initial model $y_i = \beta_1 \cdot T_i + \beta_2 X_i + \epsilon_i$ where T_i is the introduction of the technology and X_i is a set of controls. The method decomposes the effect of the treatment into two parts:

$$\frac{dy}{dT} = \frac{\partial y}{\partial M} \frac{\partial M}{\partial T} + R \quad (\text{A.1})$$

where M is the mediator –*Number of Tests*–. From (A.1) it is possible to isolate R given information on all other three elements. To do this, we substitute $\frac{dy}{dT}$ by the $\hat{\beta}$ from (2) where length of stay is the dependent variable. Secondly, we estimate $\hat{\beta}_{inter} = \frac{\partial M}{\partial T}$ from again regressing (2) but now having the mediator variable as the dependent variable. Lastly, we add the mediator M as an additional independent variable in (2) and obtain its estimated coefficient $\hat{\beta}_{med}$, which we take as an approximation to $\frac{\partial y}{\partial M}$. We can then define the ratio of mediator j as:

$$\frac{\hat{\beta}_{med(j)} \times \hat{\beta}_{inter(j)}}{\hat{\beta}}$$

We find that the mediator ratio is 48%, suggesting that around half of the treatment effect is due to physicians choosing to acquire different information (in this case, by ordering fewer tests). This implies that around 52% of the effect is independent of the mediating variable – which may be driven by other mechanisms such as the type of tests ordered, as well as the ability to learn better from the information acquired.

Figures and Tables

TABLE A.1: SUMMARY STATISTICS BY TIME AND WARD

Before														
	Prepaid Ward							Non Prepaid Ward						
	Mean	SD	p10	p25	p50	p75	p90	Mean	SD	p10	p25	p50	p75	p90
Length of Stay (Days)	.4	.55	.03	.07	.17	.39	1.27	.79	.78	.04	.11	.39	1.74	2
Number of Tests	4.68	6.58	0	0	2	8	12	10.26	14.2	0	1	6	14	28
Total Cost	2.62	12.97	.1	.18	.43	.91	5	11.41	35.89	.1	.28	.73	7.09	28.96
Hospitalization	.15	.36	0	0	0	0	1	.41	.49	0	0	0	1	1
30-Day Return	.12	.33	0	0	0	0	1	.13	.34	66	0	0	0	1

After														
	Prepaid Ward							Non Prepaid Ward						
	Mean	SD	p10	p25	p50	p75	p90	Mean	SD	p10	p25	p50	p75	p90
Length of Stay (Days)	.38	.52	.03	.06	.16	.39	1.16	.95	.77	.09	.23	.75	2	2
Number of Tests	4.49	6.34	0	0	1	7	12	11.68	13.64	0	1	8	16	29
Total Cost	2.66	12.03	.1	.17	.41	.88	5.18	14.57	38.11	.14	.4	2.03	10.98	38.88
Hospitalization	.14	.35	0	0	0	0	1	.52	.5	0	0	1	1	1
30-Day Return	.12	.32	0	0	0	0	1	.11	.32	0	0	0	0	1

Note: This table displays summary statistics for the main outcomes. Length of stay is the time between triage and the departure of the patient from the ED (i.e. discharge or hospital admission). Number of tests is the number of laboratory tests ordered during the patient stay in the ED. Total charges (defined as the log of the amount in Million of Colombian Pesos (COP) charged by the hospital to the payer for each episode). The hospital admission dummy takes value one if the patient was admitted to the hospital instead of discharged home. The 30-day return dummy takes value one if the patient returns to the ED within 30 days. The sample includes between March and June 2022 (defined as before) and from July to October 2022 (defined as after).

TABLE A.2: SUMMARY STATISTICS BY TIME AND WARD – CONTROLS

Before														
	Private Ward							Non Private Ward						
	Mean	SD	p10	p25	p50	p75	p90	Mean	SD	p10	p25	p50	p75	p90
Male Patient	.38	.49	0	0	0	1	1	.42	.49	0	0	0	1	1
Patient Age	46.67	19.05	22	31	44	60	74	47.26	19.72	24	30	43	63	76
Triage 1	.01	.07	0	0	0	0	0	.1	.3	0	0	0	0	1
Triage 2	.04	.21	0	0	0	0	0	.64	.48	0	0	1	1	1
Triage 3	.63	.48	0	0	1	1	1	.23	.42	0	0	0	0	1
Triage 4	.31	.46	0	0	0	1	1	.02	.15	0	0	0	0	0

After														
	Private Ward							Non Private Ward						
	Mean	SD	p10	p25	p50	p75	p90	Mean	SD	p10	p25	p50	p75	p90
Male Patient	.39	.49	0	0	0	1	1	.49	.5	0	0	0	1	1
Patient Age	47.18	19.05	23	31	45	61	75	50.53	20.16	24	32	50	67	78
Triage 1	.01	.08	0	0	0	0	0	.13	.34	0	0	0	0	1
Triage 2	.05	.21	0	0	0	0	0	.66	.47	0	0	1	1	1
Triage 3	.69	.46	0	0	1	1	1	.19	.39	0	0	0	0	1
Triage 4	.25	.43	0	0	0	0	1	.02	.14	0	0	0	0	0

Note: This table displays summary statistics for control variables. Male patient is a dummy equal to one when the patient is male and zero otherwise, the patient's age, and a dummy for each one of the four first triages. The sample includes between March and June 2022 (defined as after) and from July to October 2022 (defined as after).

TABLE A.3: BASELINE FINDINGS

Dependent Variable = Log Length of Stay				
	(1)	(2)	(3)	(4)
	DiD	Placebo DiD	Baseline DiDiD	Baseline DiDiD
Year =	2022	2019	2019&2022	2019&2022
Post June X Private Ward	-.247 (.043)	-.115 (.026)	-.115 (.026)	-.095 (.019)
Post June X Private Ward X Year 2022			-.133 (.051)	-.051 (.033)
Log Number of Tests				.777 (.011)
Patient Controls	Yes	Yes	Yes	Yes
Doctor Fixed Effects	Yes	Yes	Yes	Yes
Insurance Status Fixed Effects	Yes	Yes	Yes	Yes
Date X Hour Fixed Effects	Yes	Yes	Yes	Yes
Interactions with 2022 dummy	No	No	Yes	Yes
Observations	29,108	32,492	61,600	61,600

Note: This Table displays estimates of regressions of a case's length of stay in the ED on the period during which the technology was introduced (i.e. after June), interacted with the ward in which it was introduced (i.e. private ward). The unit of observation is a case i arriving at the ED. The estimating equation in Columns 1 and 2 is:

$$y_i = \beta(\text{Private}_{w(i)} \times \text{Post}_{t(i)}) + \alpha_{d(i)} + \theta_{in(i)} + \pi_{t(i)} + \gamma' \mathbf{X}_i + \epsilon_i$$

where w indexes the ward to which the patient is assigned, t indexes the exact hour (i.e. date/hour of day combination) in which the patient arrived and d indexes the physician to which the patient was assigned. Column 1 (2) shows the results of running this regression only for 2022 (2019). The main independent variable of interest is the interaction between being assigned to the private ward and arriving in the month of July or after. The model controls for insurance status (which subsumes the assigned ward), physician and hour-fixed effects, as well as patient controls (age, gender, dummies for the main diagnosis, as well as dummies for extreme vital signs markers upon admission). In Column 2 we repeat the Column 1 exercise on the placebo sample of 2019. In Columns 3 and, 4 the sample includes cases from both 2019 and 2022. The main independent variable of interest is the triple interaction between the private ward, the after-June dummy, and the 2022 dummy. The model controls for the interactions between all the controls and the year 2022 dummy. Standard errors are clustered at the physician level.

TABLE A.4: LENGTH OF STAY BY TEST CRITICALITY

Log. Length of Stay			
	Full	Critic	Non-Critic
Post June X Private Ward X 2022	-.011 (.027)	.161 (.123)	-.019 (.026)
Coef. Diffs. P-value	—		.002
Month Fixed Effects X Year 2022	Yes	Yes	Yes
Year Fixed Effects X Year 2022	Yes	Yes	Yes
Insurance Fixed Effects X Year 2022	Yes	Yes	Yes
Patient Controls X Year 2022	Yes	Yes	Yes
Observations	25,582	1,114	24,466

Note: This Table displays estimates of regressions of a case's length of stay in the ED on the period during which the technology was introduced (i.e. after June), interacted with the ward in which it was introduced (i.e. private ward) and with the year 2022 dummy. The unit of observation is a case i arriving at the ED. The estimating equation is:

$$y_i = \beta(Private_{w(i)} \times Post_{t(i)} \times 2022_{t(i)}) + \alpha_{d(i)} + \theta_{w(i)} + \pi_{t(i)} + \gamma'(\mathbf{X}_i \times 2022_{t(i)}) + \epsilon_i$$

where w indexes the ward to which the patient is assigned, t indexes the month-year combination in which the patient arrived and d indexes the physician to which the patient was assigned. The main independent variable of interest is the interaction between being assigned to the private ward, arriving in the months between July and October, and arriving in 2022. The model controls for insurance status (which subsumes the assigned ward), month and year fixed effects, as well as patient controls (age, gender, dummies for the main diagnosis, as well as dummies for extreme vital signs markers upon admission). The model controls for month, year and insurance status (which subsumes the assigned ward), as well as patient controls (age, gender, dummies for the main diagnosis, as well as dummies for extreme vital signs markers upon admission) interacted with the 2022 dummy. Standard errors are clustered at the physician level. The columns differ in the sample used, based on the analysis of laboratory test results, which include both the test outcome and its lower and upper critical thresholds. A test is defined as critical if its result falls below the lower threshold or above the upper range. A dummy variable was created, equal to 1 if an episode has at least one critical test and 0 if it has no critical tests. Full columns include all episodes with tests for which both the result and criticality range information are available. Critic columns include only episodes with at least one test whose result can be classified as critical, while Non-critic columns include episodes without any tests classified as critical. Based on this classification, the main regression in Column 6 of Table 2 uses the full sample of episodes, with a total of 61,600 observations, while the regression in Column 1, which relies on episodes with test results, includes 25,582 observations, representing 40% of the full sample.

TABLE A.5: LENGTH OF STAY WITH AND WITHOUT TESTS

	(1) All Episodes	(2) Without Tests	(3) With Tests
Post June X Private Ward X Year 2022	-.133 (.051)	-.072 (.067)	-.156 (.052)
Post June X Private Ward	Yes	Yes	Yes
Patient Controls X Year 2022	Yes	Yes	Yes
Doctor Fixed Effects X Year 2022	Yes	Yes	Yes
Insurance Status Fixed Effects X Year 2022	Yes	Yes	Yes
Date X Hour Fixed Effects	Yes	Yes	Yes
Post June X Year 2022	No	No	No
Private Ward X Year 2022	No	No	No
Observations	61,600	20,892	37,270

Note: This table displays estimates of regressions of a case's length of stay in the ED on the period during which the informational dashboard was introduced (i.e. post 18 June), interacted with the ward in which it was introduced (i.e. private ward) and with the year 2022 dummy. The unit of observation is a case i arriving at the ED. The estimating equation is:

$$y_i = \beta(Private_{w(i)} \times Post_{t(i)} \times 2022_{t(i)}) + \alpha_{d(i)} + \theta_{w(i)} + \pi_{t(i)} + \gamma'(\mathbf{X}_i \times 2022_{t(i)}) + \epsilon_i$$

where w indexes the ward to which the patient is assigned, t indexes the exact hour (i.e. date/hour of day combination) in which the patient arrived and d indexes the physician to which the patient was assigned. The main independent variable of interest is the interaction between being assigned to the private ward, arriving in the months between July and October, and arriving in 2022. The model controls for insurance status (which subsumes the assigned ward), physician and hour fixed effects, as well as patient controls (age, gender, dummies for the main diagnosis, as well as dummies for extreme vital signs markers upon admission) interacted with the year 2022 dummy. Column 1 includes all episodes. Column 2 only includes episodes in which no tests were requested, while Column 3 only includes episodes in which there was at least one requested test. Standard errors are clustered at the physician level.

TABLE A.6: THE IMPACT ON CASES REQUIRING TESTS

Dependent Variable =	(1) Log. Length of Stay	(2) Log. Number of Tests	(3) Total Cost	(4) Hospital.	(5) 30-Day Return
Post June X Diag. with Test	-.203 (.05)	-.197 (.036)	-.173 (.062)	-.04 (.017)	.001 (.017)
Patient Controls X Year 2022	Yes	Yes	Yes	Yes	Yes
Dr. Fixed Effects X Year 2022	Yes	Yes	Yes	Yes	Yes
Ins. Fixed Effects X Year 2022	Yes	Yes	Yes	Yes	Yes
Date X Hour Fixed Effects	Yes	Yes	Yes	Yes	Yes
Observations	35,712	36,100	36,066	36,100	36,100

Note: This table displays estimates of regressions of a case's different medical outcomes in the ED on the period during which the technology was introduced (i.e. after June), interacted with a dummy equal to 1 for diagnosis with tests (defined as a diagnosis that received at least one test in the first semester of 2022 (excluding June)) and 0 otherwise. The estimating equation is:

$$y_i = \beta(\times Post_{t(i)} \times DiagTest_i) + \alpha_{d(i)} + \theta_{w(i)} + \pi_{t(i)} + \gamma'(\mathbf{X}_i) + \epsilon_i$$

The model controls for insurance status (which subsumes the assigned ward), physician and hour-fixed effects, as well as patient controls (age, gender, dummies for the main diagnosis, as well as dummies for extreme vital signs markers upon admission). In column 1, the dependent variable is the case's length of stay in the ED. In column 2, the dependent variable is the logarithm of the number of tests. In column 3, the dependent variable is the total charge (defined as the log of the amount in Colombian Pesos (COP) charged by the hospital to the payer for each episode). In Column 4, the dependent variable is a dummy if the patient is hospitalized. The last column, the dependent variable is a dummy that takes the value of 1 if the patient returns to the ED within a 30-day period. Standard errors are clustered at the physician level.

TABLE A.7: LENGTH OF STAY BY WHETHER TESTS WERE HISTORICALLY REQUIRED

Dependent Variable =	Log. Length of Stay		
	Full Sample	ICD with Tests	ICD without Tests
Post June X Private Ward X 2022	-.123 (.028)	-.139 (.05)	-.059 (.038)
Coef. Diff. P-value	—		.007
Patient Controls X Year 2022	Yes	Yes	Yes
Dr. Fixed Effects X Year 2022	Yes	Yes	Yes
Ins. Fixed Effects X Year 2022	Yes	Yes	Yes
Date X Hour Fixed Effects	Yes	Yes	Yes
Observations	61,612	26,924	31,162

This Table displays estimates of regressions of a case's length of stay in the ED on the period during which the technology was introduced (i.e. after June), interacted with the ward in which it was introduced (i.e. private ward) and with the year 2022 dummy. The unit of observation is a case i arriving at the ED. The estimating equation is:

$$y_i = \beta(Private_{w(i)} \times Post_{t(i)} \times 2022_{t(i)}) + \alpha_{d(i)} + \theta_{w(i)} + \pi_{t(i)} + \gamma'(\mathbf{X}_i \times 2022_{t(i)}) + \epsilon_i$$

where w indexes the ward to which the patient is assigned, t indexes the exact hour (i.e. date/hour of day combination) in which the patient arrived and d indexes the physician to which the patient was assigned. The main independent variable of interest is the interaction between being assigned to the private ward, arriving in the months between July and October, and arriving in 2022. The model controls for insurance status (which subsumes the assigned ward), physician and hour-fixed effects, as well as patient controls (age, gender, dummies for the main diagnosis, as well as dummies for extreme vital signs markers upon admission). The dependent variables are the log length of stay, total charge (defined as the log of the amount in Colombian Pesos (COP) charged by the hospital to the payer for each episode), a dummy if the patient is hospitalized, a dummy that takes the value of 1 if the patient returns to the ED within a 30-days period and log number of tests. All the controls are interacted with the year 2022 dummy. Results are presented for the full sample, as well as sample cuts based on the median doctor age in 2019. Certain conditions, such as fractures, generally do not require diagnostic tests, while others, like infections or metabolic disorders, often do. To differentiate these cases, ICD-10 groups were classified based on their typical need for tests. Conditions requiring tests include infectious diseases (A00-B99), tumors (C00-D48), blood and immune disorders (D50-D89), endocrine and metabolic diseases (E00-E90), nervous system disorders (G00-G99), cardiovascular diseases (I00-I99), respiratory diseases (J00-J99), digestive diseases (K00-K93), genitourinary diseases (N00-N99), pregnancy and perinatal conditions (O00-O99, P00-P96), and congenital malformations (Q00-Q99), as these often require blood tests, imaging, or biopsies for diagnosis. Conversely, conditions that generally do not require tests include mental and behavioral disorders (F00-F99), eye diseases (H00-H59), ear and mastoid disorders (H60-H95), skin and subcutaneous tissue diseases (L00-L99), musculoskeletal disorders (M00-M99), general symptoms and abnormal clinical findings (R00-R99), injuries and external causes (S00-T98, V00-Y99), and health-related factors and contact with health services (Z00-Z99), which are primarily diagnosed through clinical evaluation. All ICD-10 codes starting with 'U' (U00-U99) have been classified as requiring diagnostic tests. While some codes in this category are used for statistical reporting or administrative purposes, the majority represent emerging infectious diseases, antimicrobial resistance, or other conditions that typically require laboratory confirmation or imaging. To maintain consistency, all 'U' codes have been grouped under conditions that require testing.

TABLE A.8: EVIDENCE ON PEER EFFECTS

	Log. Length of Stay		Log. Number of Tests		Total Cost		Hospitalization		30-Day Return	
	Number of Drs.		Number of Drs.		Number of Drs.		Number of Drs.		Number of Drs.	
	1	2+	1	2+	1	2+	1	2+	1	2+
Post June X Private Ward X 2022	-.251 (.021)	-.068 (.056)	-3.617 (.294)	-.294 (1.306)	-.605 (.055)	-.055 (.131)	-.174 (.014)	-.02 (.04)	.027 (.018)	-.018 (.036)
Patient Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Doctor Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Insurance Status Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date X Hour Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	46,629	12,829	46,629	12,829	46,562	12,796	46,629	12,829	46,629	12,829

Note: This Table displays estimates of regressions of a case's number of physicians working in an episode on patient outcomes on the period during which the informational dashboard was introduced (i.e. after June), interacted with the ward in which it was introduced (i.e. private ward) and with the year 2022 dummy. To define the number of physicians working in an hour-ward-episode, we count the total number of physicians that are directly treating patients, that are requesting tests for patients, or that close any ED episode. The estimating equation is:

$$y_i = \beta(Private_{w(i)} \times Post_{t(i)} \times 2022_{t(i)}) + \alpha_{d(i)} + \theta_{w(i)} + \pi_{t(i)} + \gamma'(\mathbf{X}_i \times 2022_{t(i)}) + \epsilon_i$$

where w indexes the ward to which the patient is assigned, t indexes the exact hour (i.e. date-hour of day combination) in which the patient arrived and d indexes the physician to which the patient was assigned. The main independent variable of interest is the interaction between being assigned to the private ward, arriving in the months between July and October, and arriving in 2022. The model controls for insurance status (which subsumes the assigned ward), physician and hour fixed effects, as well as patient controls (age, gender, dummies for the main diagnosis, as well as dummies for extreme vital signs markers upon admission) interacted with the year 2022 dummy. Sample changes for each column depending on the number of physicians working on an episode (treatment physician and other physicians who request tests). Standard errors are clustered at the physician level.

TABLE A.9: LENGTH OF TEST PROCESSING

	(1)	(2)	(3)	(4)	(5)
Post June X Private Ward X Year 2022	-.001 (.035)	.005 (.034)	.009 (.034)	.045 (.033)	-.026 (.031)
Patient Controls X Year 2022	Yes	No	Yes	Yes	Yes
Doctor Fixed Effects	Yes	Yes	No	Yes	Yes
Insurance Status Fixed Effects	Yes	Yes	Yes	No	Yes
Date X Hour Fixed Effects	Yes	Yes	Yes	Yes	No
Observations	37,555	37,555	37,701	37,555	39,507

Note: This Table displays estimates of regressions of a case's the total time that the test took to be delivered in the laboratory department on the period during which the technology was introduced (i.e. after June), interacted with the ward in which it was introduced (i.e. private ward) interacted with the year 2022. The unit of observation is a case i arriving at the ED. The estimating equation is:

$$y_i = \beta(Private_{w(i)} \times Post_{t(i)} \times 2022_{t(i)}) + \alpha_{d(i)} + \theta_{w(i)} + \pi_{t(i)} + \gamma'(\mathbf{X}_i \times 2022_{t(i)}) + \epsilon_i$$

where w indexes the ward to which the patient is assigned, t indexes the exact hour (i.e. date-hour of day combination) in which the patient arrived, and d indexes the physician to which the patient was assigned. The main independent variable of interest is the interaction between being assigned to the private ward, arriving in the months between July and October, and arriving in 2022. The model controls for insurance status (which subsumes the assigned ward), physician and hour-fixed effects, as well as patient controls (age, gender, dummies for the main diagnosis, as well as dummies for extreme vital signs markers upon admission) interacted with the 2022 dummy. Standard errors are clustered at the physician level.

TABLE A.10: EVIDENCE ON COGNITIVE MECHANISMS

PANEL A: EVIDENCE CONSISTENT WITH THE LEARNING MECHANISM

	Log. Length of Stay		Log. Number of Tests		Total Cost		Hospitalization		30-Day Return	
	Busy	Non-Busy	Busy	Non-Busy	Busy	Non-Busy	Busy	Non-Busy	Busy	Non-Busy
Post June X Private Ward X Year 2022	-.533 (.211)	-.132 (.062)	-.497 (.154)	-.108 (.043)	-.438 (.242)	-.328 (.077)	-.144 (.051)	-.057 (.017)	.054 (.063)	.001 (.015)
Patient Controls X Year 2022	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Doctor Fixed Effects X Year 2022	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Insurance Status Fixed Effects X Year 2022	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date X Hour Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	17,889	43,064	18,190	43,839	18,151	43,773	18,190	43,839	18,190	43,839

PANEL B: LENGTH OF STAY BY DIAGNOSIS CLASSIFICATION

	(1) All Diagnosis	(2) Common (20/80 Split) Diagnosis	(3) Uncommon (20/80 Split) Diagnosis	(4) Common (50/50 Split) Diagnosis	(5) Uncommon (50/50 Split) Diagnosis
Post June X Private Ward X Year 2022	-.133 (.051)	.019 (.13)	-.378 (.079)	-.13 (.082)	-.427 (.092)
Patient Controls X Year 2022	Yes	Yes	Yes	Yes	Yes
Doctor Fixed Effects X Year 2022	Yes	Yes	Yes	Yes	Yes
Insurance Status Fixed Effects X Year 2022	Yes	Yes	Yes	Yes	Yes
Date X Hour Fixed Effects	Yes	Yes	Yes	Yes	Yes
Observations	61,600	9,977	47,651	24,385	33,735

Note: Panel A from this Table displays estimates of regressions of a case's log length of stay, log of the number of tests, total charges, hospitalization and 30-day return in the ED on the period during which the informational dashboard was introduced (i.e. after June), interacted with the ward in which it was introduced (i.e. private ward) and with the year 2022 dummy. This table shows results for different samples according to busy or non-busy hours. The percentile 75 for the patient-physician ratio is near 2 patients per physician. So, busy hours are defined as hours where there are more than 2 patients per physician. The unit of observation is a case i arriving at the ED. The estimating equation is:

$$y_i = \beta(Private_{w(i)} \times Post_{t(i)} \times 2022_{t(i)}) + \alpha_{d(i)} + \theta_{w(i)} + \pi_{t(i)} + \gamma'(\mathbf{X}_i \times 2022_{t(i)}) + \epsilon_i$$

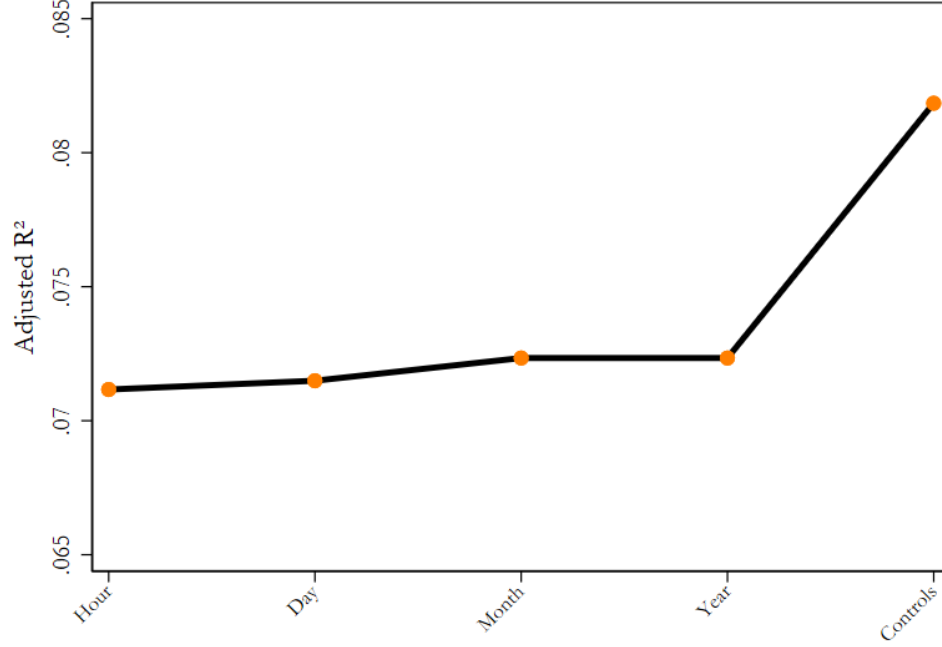
where w indexes the ward to which the patient is assigned, t indexes the exact hour (i.e. date/hour of day combination) in which the patient arrived and d indexes the physician to which the patient was assigned. The main independent variable of interest is the interaction between being assigned to the private ward, arriving in the months between July and October, and arriving in 2022. The model controls for insurance status (which subsumes the assigned ward), physician and hour fixed effects, as well as patient controls (age, gender, dummies for the main diagnosis, as well as dummies for extreme vital signs markers upon admission) interacted with the year 2022 dummy. In Columns 1-2, the dependent variable is the length of stay. In Columns 3-4, the dependent variable is the total charge (defined as the log of the amount in Colombian Pesos (COP) charged by the hospital to the payer for each episode). In Columns 5-6, the dependent variable is the number of tests. In Columns 7-8, the dependent variable is a dummy if the patient is hospitalized. In Columns 9-10, the dependent variable is a dummy that takes the value of 1 if the patient returns to the ED within a 30-days period. Within a pair of columns, the first column of each dependent variable shows the busy sample while the second column shows the non-busy sample. Standard errors are clustered at the physician level.

Panel B from this table displays estimates of regressions of a case's log. length of stay. The estimating equation is:

$$y_i = \beta(Private_{w(i)} \times Post_{t(i)} \times 2022_{t(i)}) + \theta_{w(i)} + \pi_{t(i)} + \gamma'(\mathbf{X}_i \times 2022_{t(i)}) + \epsilon_i$$

where w indexes the ward to which the patient is assigned, t indexes the exact hour (i.e. date/hour of day combination) in which the patient arrived and d indexes the physician to which the patient was assigned. The main dependent variable of interest is the length of stay. The model controls for insurance status (which subsumes the assigned ward), physician and hour fixed effects, as well as patient controls (age, gender, dummies for the main diagnosis, as well as dummies for extreme vital signs markers upon admission) interacted with the year 2022 dummy. Standard errors are clustered at the physician level. Column 1 includes episodes of all the diagnoses. Column 2 includes only the common diagnosis and Column 3 includes the uncommon diagnosis, both based on a 20%–80% distribution cut. Column 4 includes only the common diagnosis and Column 5 includes the uncommon diagnosis, both based on a 50%–50% distribution cut.

FIGURE A.1: INCREMENTAL EXPLANATORY POWER OF OBSERVABLES IN TEST PROCESSING TIME



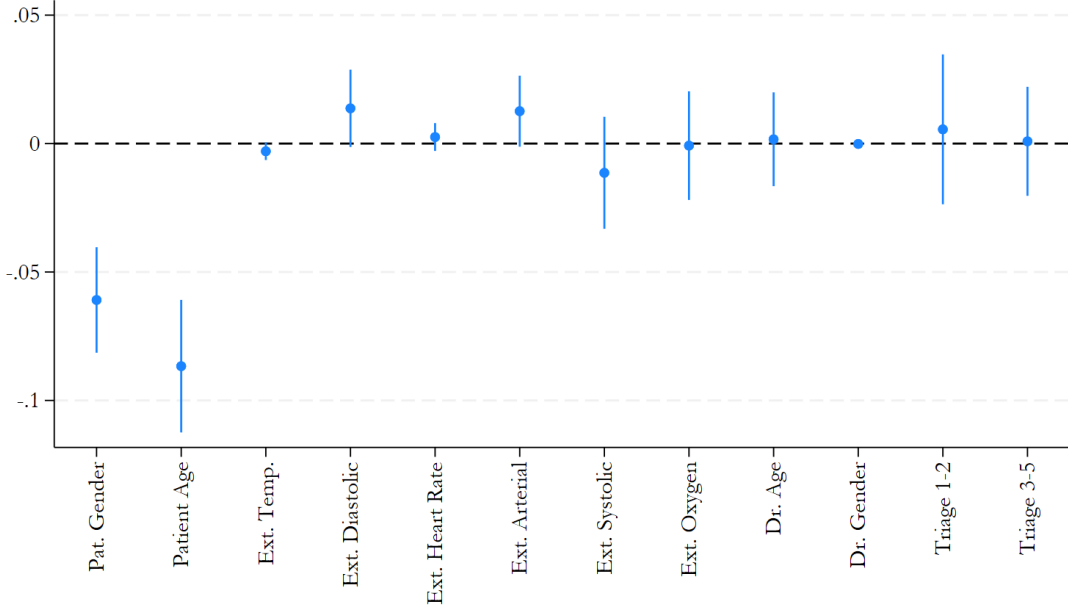
Note: This table presents the adjusted R^2 values for regressions analyzing how observable variables influence the length of test processing. The key explanatory variable is the interaction between the period in which the technology was introduced (i.e., after June), the ward where it was implemented (i.e., private ward), and a dummy for the year 2022. The unit of observation is a case i arriving at the ED, and the estimation equation for the most comprehensive specification (Controls) includes both patient-level observables (age, gender, dummies for the main diagnosis, as well as dummies for extreme vital sign markers upon admission) and time-related observables associated with the patient's admission date. The estimation equation is:

$$y_i = \beta(Private_{w(i)} \times Post \times 2022) + \gamma_{j(i)} + \alpha_{w(i)} + \theta_{d(i)} + \beta_{t(i)} + \mathbf{X}_i + \epsilon_i$$

where j indexes the hour, w the day, d the month, and t the year to which the patient is assigned. The table presents results from a stepwise inclusion of fixed effects:

- Hour: Includes only hour fixed effects, without patient controls.
- Day: Adds day fixed effects.
- Month: Includes month fixed effects in addition to hour and day fixed effects.
- Year: Incorporates all fixed effects (Hour, Day, Month, and Year) but excludes patient controls.
- Controls: Includes all fixed effects and patient controls, allowing us to assess the contribution of observable patient characteristics to the variation in test processing time.

FIGURE A.2: EXOGENEITY



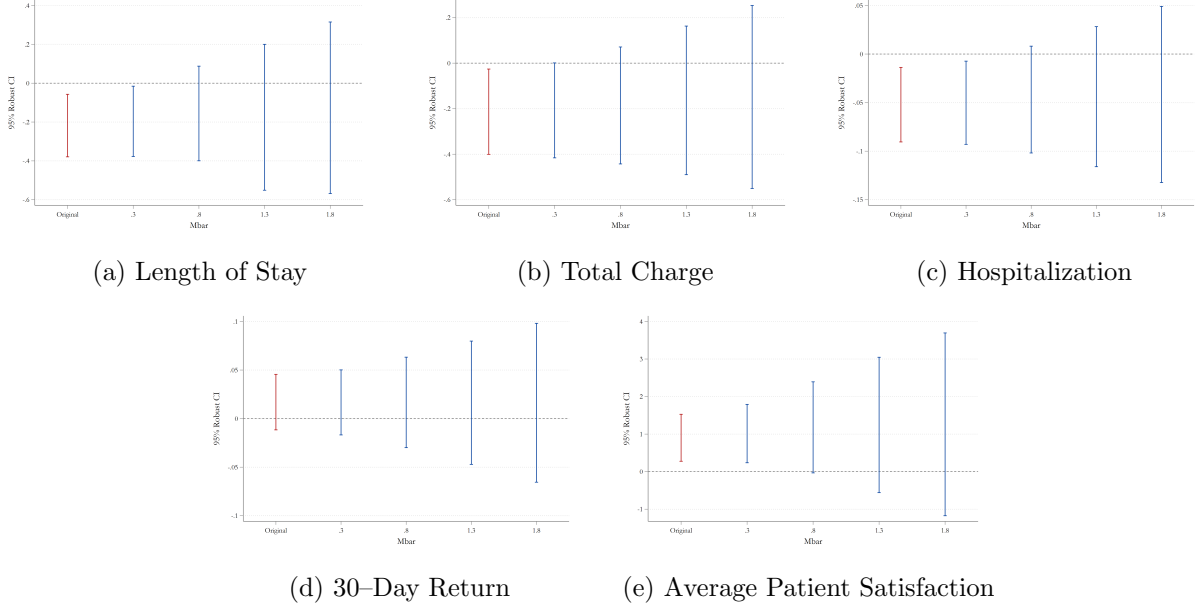
The adjusted R^2 measures the proportion of variance in test processing time explained by the model, penalizing for the inclusion of additional regressors to ensure a more accurate assessment of explanatory power. Comparing specifications helps to quantify the incremental explanatory power of including patient characteristics and time fixed effects. Higher adjusted R^2 values in more comprehensive specifications indicate that observable factors play a meaningful role in explaining differences in test processing time, beyond random variation.

Note: This figure displays coefficients of different regressions for the following dependent variables y_i : Patient Gender (dummy defined as 1 for men and 0 for women), Patient Age dummy defined as 1 for patients whose age is above the median and 0 otherwise, Extreme Vital Signs (diastolic pressure, heart rate, arterial pressure, systolic pressure and oxygen saturation), Dr. Age dummy defined as 1 for physicians whose age is above the median and 0 otherwise, Dr. Gender (dummy defined as 1 for men and 0 for women), low triage dummy defined as 1 for triages 1 and 2 and 0 otherwise. The estimating equation is:

$$y_i = \beta(Private_{w(i)} \times Post_{t(i)} \times 2022_{t(i)}) + \theta_{w(i)} + \pi_{t(i)} + \gamma'(\mathbf{X}_i \times 2022_{t(i)}) + \epsilon_i$$

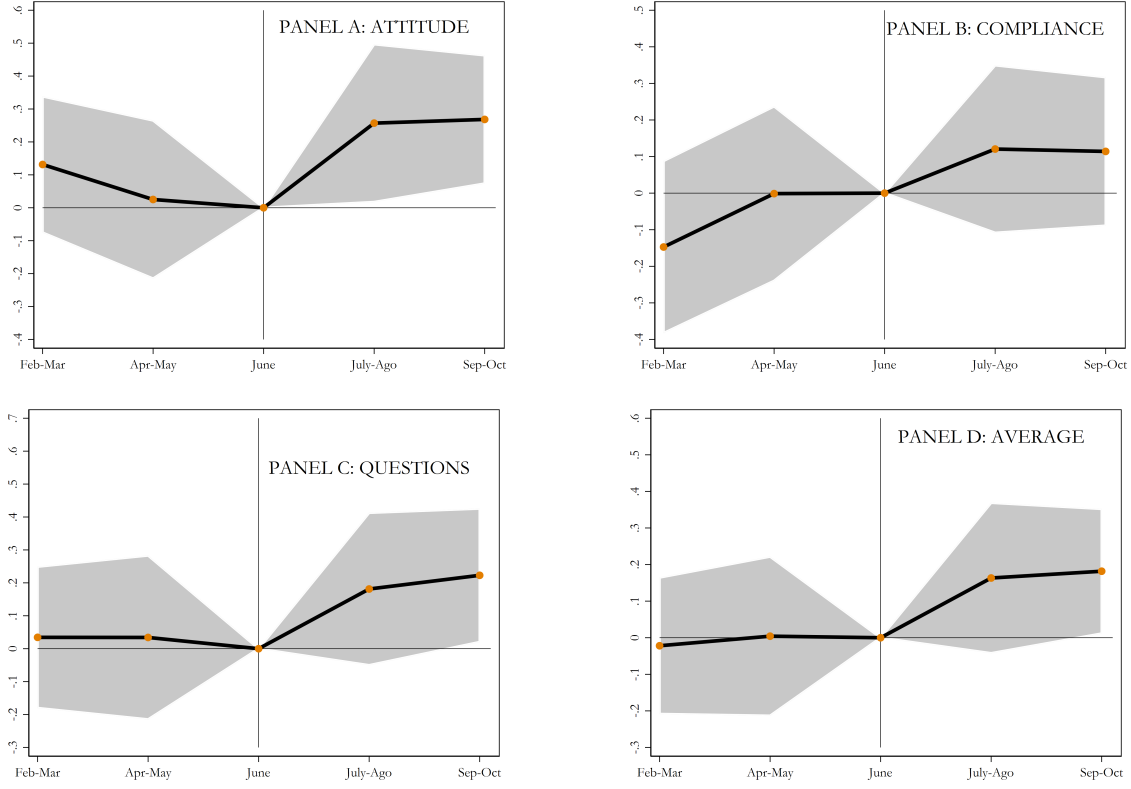
where w indexes the ward to which the patient is assigned, t indexes the exact hour (i.e. date-hour of day combination) in which the patient arrived and d indexes the physician to which the patient was assigned. The model controls for insurance status (which subsumes the assigned ward), physician and hour fixed effects, as well as patient controls (age, gender, dummies for the main diagnosis, as well as dummies for extreme vital signs markers upon admission) interacted with the year 2022 dummy. When the variable is the dependent variable, it is excluded from the control list. For instance, for the first variable, the dependent variable is patient gender and in this case, is excluded from the list $\mathbf{X}_i$. Standard errors are clustered at the physician level.

FIGURE A.3: HONEST DiD



Note: These figures display coefficients for sensitivity analysis before and after the treatment. It shows a robust confidence interval for different values of proportional violation of parallel trends assumption. The value equal to 1, for instance, imposes that the post-treatment violation of parallel trends is no longer than the worst pre-treatment violation of parallel trends (between consecutive periods). Likewise, a value of 2 implies that the post-treatment violation of parallel trends is no more than twice that in the pre-treatment period. Where the coefficients cross the zero line (breakdown value), it means that the significant result is robust to allowing for violations of parallel trends up to the value and as big as the max violation in the pre-treatment period.

FIGURE A.4: LEADS AND LAGS EVIDENCE DOUBLE-DIF

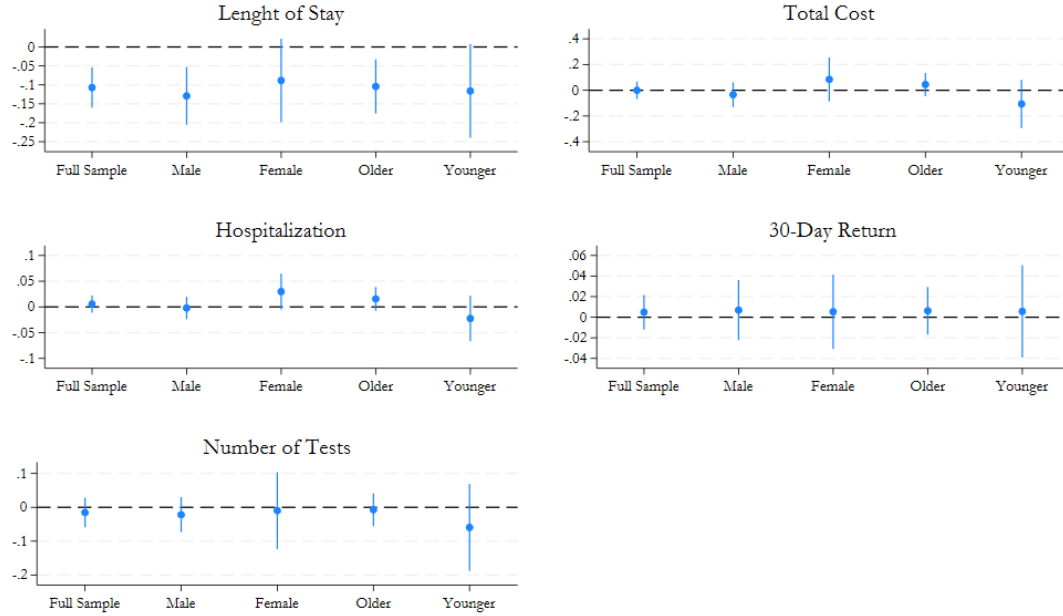


This Figure displays dynamic estimates of regressions of a case's survey evaluation on the period during which the technology was introduced (i.e. after June), interacted with the ward in which it was introduced (i.e. private ward). The unit of observation is a case i arriving at the ED. This figure displays the coefficients β_t from estimating:

$$y_i = \sum_{t=2-3,4-5}^{7-8,9-10} \beta_t (Private_{w(i)} \times Month_{t(i)}) + \alpha_{d(i)} + \theta_{w(i)} + \pi_{t(i)} + \gamma' \mathbf{X}_i + \epsilon_i$$

where w indexes the ward to which the patient is assigned, t indexes the exact hour (i.e. date/hour of day combination) in which the patient arrived, and d indexes the physician to which the patient was assigned. The main independent variable of interest is the interaction between being assigned to the private ward and arriving after June. The model controls for insurance status (which subsumes the assigned ward), physician and hour-fixed effects, as well as patient controls (age, gender, and health markers upon admission). Standard errors are clustered at the physician level.

FIGURE A.5: HETEROGENEITY BY DOCTOR GENDER AND AGE



Note: This figure presents coefficient estimates from regressions of a case's different medical outcomes in the ED on the period during which the technology was introduced (i.e. after June), interacted with a dummy equal to 1 for diagnosis without tests in the pre-period (before June 2022) and 0 otherwise. The estimating equation is:

$$y_i = \beta(Post_{t(i)} \times Private_{w(i)} \times 2022_{t(i)}) + \alpha_{d(i)} + \theta_{w(i)} + \pi_{t(i)} + \gamma'(\mathbf{X}_i \times 2022_{t(i)}) + \epsilon_i$$

where w indexes the ward to which the patient is assigned, t indexes the exact hour (i.e., date/hour of day combination) in which the patient arrived, and d indexes the physician to which the patient was assigned. The main independent variable of interest is the interaction between being assigned to the private ward, arriving in the months between July and October, and arriving in 2022. The model controls for insurance status (which subsumes the assigned ward), physician and hour-fixed effects, as well as patient controls (age, gender, dummies for the main diagnosis, and dummies for extreme vital signs markers upon admission). All the patient controls are interacted with the year 2022 dummy. Results are presented for the full sample, as well as for sample splits based on doctor gender and age (young and old doctors were classified using age data from the years 2019 and 2022. For each doctor, the average of their recorded age across these two years was calculated. Based on this average, the 50th percentile (median) was computed, and doctors whose average age was above the median were classified as “Old”, while those below the median were classified as “Young”).